

RESEARCH ARTICLE

Hybrid Deep Learning for Human Fall Detection: A Synergistic Approach Using YOLOv8 and Time-Space Transformers

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ABSTRACT Falls are a major concern, particularly for elderly individuals and vulnerable populations, often leading to severe injuries if not detected promptly. Traditional sensor-based fall detection methods suffer from limitations such as discomfort, maintenance challenges, and susceptibility to false readings. To address these issues, we propose a hybrid deep learning-based system that leverages video-based techniques for accurate and efficient fall detection. Our approach integrates YOLOv8 for real-time human detection with Time-Space Transformers for temporal motion analysis, ensuring robust recognition of fall incidents while minimizing false positives and negatives. The proposed system is evaluated on two benchmark datasets, CUCAFall and DiverseFALL10500, which contain diverse and challenging environmental scenarios. Experimental results demonstrate that our model outperforms existing state-of-the-art methods, achieving a mean Average Precision (mAP) of 0.9955 on CUCAFall and 0.950 on DiverseFALL10500, along with F1-scores exceeding 0.998 for all classes in certain configurations. These results confirm the model's ability to distinguish fall events from normal activities accurately. Furthermore, the system maintains computational efficiency, making it suitable for real-time deployment in healthcare and smart home environments. By enhancing spatial and temporal awareness, our hybrid approach significantly improves fall detection reliability, ensuring timely intervention and enhancing safety for at-risk individuals.

INDEX TERMS Deep fall detection, CUCAFall, DiverseFALL10500, human detection, time-space transformers, YOLOv8.

I. INTRODUCTION

Human fall detection has become essential to healthcare monitoring, particularly for elderly individuals and vulnerable populations [1]. Falls are a leading cause of injury and can result in severe health complications if not detected promptly [2]. Early detection and response are critical in minimizing injury severity [3], which makes automatic fall detection systems highly valuable in various environments such as hospitals [4], care homes, and even smart homes.

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Traditional fall detection methods often rely on wearable sensors, but they can be intrusive, uncomfortable, and prone to limitations such as battery life and sensor positioning [5]. Therefore, video-based fall detection systems have emerged as a non-intrusive, scalable, and effective solution for real-time monitoring [2]. The primary challenge in video-based fall detection is accurately identifying fall incidents while minimizing false positives and negatives [6]. Falls can vary significantly in how they manifest, and often, the difference between a fall and a regular activity can be subtle [7]. Recent studies have explored innovative techniques for fall detection, including video-based anomaly detection frameworks such

as the Magnitude-Shape (MS) Plot [8], which effectively distinguishes anomalies in crowded video scenes by analyzing magnitude and shape deviations. Another notable approach combines accelerometric data with vision-based classification [9], using exponentially Weighted Moving Average (EWMA) monitoring and Support Vector Machine (SVM)-based classification to improve detection accuracy. In addition, integrating advanced deep learning techniques has shown great promise in addressing these challenges [10]. In particular, object detection models such as YOLO (You Only Look Once) have revolutionized the ability to detect humans in video frames [11]. However, recognizing falls also requires understanding the temporal dynamics of human motion, which goes beyond frame-by-frame detection [12].

This paper proposes a hybrid approach to fall detection that combines the strengths of YOLOv8 [13], a state-of-the-art object detection model, with Time-Space Transformer networks [14], which excel at capturing the temporal evolution of human movement. YOLOv8 is used for real-time human detection, efficiently identifying individuals location and posture of individuals in each video frame [15]. Meanwhile, the Time-Space Transformer processes the sequence of frames, allowing the system to learn the temporal dependencies crucial to distinguish a fall from other activities such as sitting or bending [16]. The novelty of this paper lies in the synergy between these two powerful models. By integrating YOLOv8 and Time-Space Transformers, the proposed method enhances spatial and temporal awareness within the video sequence (i.e., images). YOLOv8 ensures high accuracy in detecting humans in individual frames [17]. At the same time, the Time-Space Transformer offers a deeper understanding of the motion patterns across frames, making it possible to detect sudden and abnormal changes in posture indicative of a fall. This hybrid approach is designed to address the limitations of single-model systems, offering improved accuracy and robustness in real-world scenarios.

This paper highlights the technical components and the practical implications of such a system. Video/image-based fall detection systems are deployed in real-time environments [18], and they must be accurate and computationally efficient to ensure prompt alerts and low latency. The integration of YOLOv8, known for its speed, and Time-Space Transformers, designed for efficient temporal modelling, ensures that the proposed system meets these real-time requirements while maintaining high detection accuracy.

A. BACKGROUND

Fall detection has become critical in healthcare monitoring and computer vision, requiring accurate and real-time detection systems to ensure timely intervention, particularly for elderly individuals and at-risk populations. Advances in object detection and motion analysis techniques have significantly enhanced the ability to detect human activities and classify fall incidents. This subsection provides an overview of the key technologies used in the proposed

system, including YOLOv8 for object detection, background subtraction for foreground extraction, and TST for temporal motion modelling. These components are vital in improving fall detection accuracy, reducing false alarms, and ensuring computational efficiency for real-time applications.

1) OBJECT DETECTION USING YOLOV8

Object detection is a critical task in computer vision, and the YOLO series of models has become one of the most popular and efficient solutions for real-time detection tasks [19], [20]. YOLOv8, the latest iteration of the YOLO family, continues this tradition by incorporating both speed and accuracy over its predecessors [21]. Object detection involves localizing objects in an image (through bounding boxes) and classifying them into predefined categories. YOLOv8 excels in this domain by framing the task as a single end-to-end process, simultaneously predicting both the bounding box coordinates and class probabilities. This efficiency makes it highly suitable for real-time applications like autonomous driving, security monitoring, and human fall detection.

The architecture of YOLOv8 builds upon the fundamental concepts of earlier versions, such as YOLOv5, but introduces several enhancements that improve detection performance [22]. The model uses a Convolutional Neural Network (CNN) backbone, typically a modified CSPDarknet, to extract hierarchical representations of features from the input image [23]. This is followed by a Path Aggregation Network (PAN) that enhances feature propagation and fusion [24], enabling the model to capture both low-level and high-level features better [23], [25]. The PAN outputs feature at multiple scales are then used to predict bounding boxes at different resolutions, addressing the need to detect objects of varying sizes. Mathematically, the output of YOLOv8 can be expressed as a set of predictions, as depicted in Equation 1:

$$\hat{y} = \{(b_x, b_y, w, h, c)\} \quad (1)$$

where b_x and b_y represent the centre coordinates of the bounding box, w and h are the width and height of the box, and c is the confidence score associated with the detected object's class. These predictions are made for each grid cell in the image, and non-maximum suppression (NMS) filters out redundant or overlapping boxes, ensuring only the most accurate predictions are retained.

YOLOv8 brings significant improvements in both the architecture and optimization techniques. One notable enhancement is introducing a more efficient anchor-free detection head, which eliminates the need for predefined anchor boxes used in earlier YOLO versions [13], [26]. Instead of relying on a set of fixed anchor boxes, YOLOv8 predicts bounding boxes dynamically, reducing the computational overhead and improving the detection of small objects. This anchor-free mechanism also simplifies the training process, as there is no need to fine-tune the anchor box sizes for specific datasets. Another improvement in YOLOv8 is using adaptive label assignment, which dynamically adjusts the contribution of positive and negative samples during

training. This allows the model to focus more on complex examples, often the most informative during the training process [13]. Formally, this can be described by the loss function in Equation 2:

$$L = L_{\text{cls}} + L_{\text{box}} + L_{\text{conf}} \quad (2)$$

where L_{cls} is the classification loss (typically cross-entropy or focal loss), L_{box} is the bounding box regression loss (commonly IoU loss), and L_{conf} is the object confidence loss. These components guide the model towards learning accurate object locations and class predictions. Bounding box regression is a critical part of YOLOv8's performance. Unlike previous versions, which relied on anchor boxes to make predictions, YOLOv8 directly predicts the offsets for each grid cell. This involves estimating the distance between the predicted box centre and the grid cell centre, along with the width and height of the box. The prediction is normalized relative to the grid cell size, ensuring that the model remains scale-invariant [13], [26]. Precisely, for a grid cell located at (i, j) , the bounding box centre can be predicted as depicted in Equation 3:

$$\hat{b}_x = i + \sigma(t_x), \quad \hat{b}_y = j + \sigma(t_y) \quad (3)$$

where t_x and t_y are the predicted offsets, and σ is the sigmoid function that ensures the predicted values are within a reasonable range. The width and height are predicted as expressed in Equation 4:

$$\hat{w} = p_w \cdot e^{t_w}, \quad \hat{h} = p_h \cdot e^{t_h} \quad (4)$$

where p_w and p_h are predefined anchor sizes, and t_w and t_h are the predicted scaling factors. This formulation allows YOLOv8 to predict bounding boxes with greater flexibility and precision.

One of the key innovations in YOLOv8 is using a Feature Pyramid Network (FPN) to handle multi-scale object detection [27]. Objects in real-world images come in various sizes, and a single resolution might not capture small objects effectively. YOLOv8 addresses this by making multiple-scale predictions, fused to form a single prediction map [28]. This allows the model to detect small objects using high-resolution feature maps, while large objects are detected using lower-resolution maps. The result is a model that can handle objects of all sizes without sacrificing speed. After the model generates multiple bounding box predictions for each grid cell, post-processing is required to filter out redundant detections. YOLOv8 uses Non-Maximum Suppression (NMS), which retains only the highest-confidence predictions while discarding overlapping boxes [15]. The NMS algorithm works by iteratively selecting the highest-confidence box and removing all boxes with an Intersection over Union (IoU) more significant than a certain threshold (typically 0.5) with the selected box [15], [29]. The result is a set of unique bounding boxes that accurately reflect the locations of objects in the image.

Training YOLOv8 involves minimizing the total loss and combining classification, localisation, and confidence loss.

The classification loss uses a cross-entropy function to penalize incorrect class predictions, while the localization loss, often computed using Intersection over Union (IoU) [30] or its variants like Generalized IoU (GIoU) [31], penalizes incorrect bounding box predictions. Additionally, the confidence loss penalizes incorrect objectness scores, which determine whether a grid cell contains an object. This multi-task learning framework ensures the model learns to detect, localize, and classify objects simultaneously. The real-time capabilities of YOLOv8 make it ideal for applications like autonomous vehicles, surveillance, and healthcare monitoring. Its ability to make predictions in real-time with minimal latency ensures that YOLOv8 can be used in critical scenarios where speed is crucial. In healthcare, for instance, YOLOv8 can be integrated with fall detection systems, where quick detection of a fall can trigger timely interventions, potentially saving lives [32], [33]. The model's lightweight architecture allows it to be deployed on edge devices with limited computational power, making it versatile for various deployment scenarios.

YOLOv8 has been shown to outperform its predecessors in both accuracy and speed benchmarks. Its ability to handle and detect small objects with high precision is beneficial in dense environments. On benchmarks like COCO (Common Objects in Context), YOLOv8 achieves high mean Average Precision (mAP) scores while maintaining real-time inference speeds [34], [35]. This balance between speed and accuracy makes YOLOv8 one of the most efficient object detectors, suitable for real-time applications with limited computational resources. While YOLOv8 significantly improved over previous versions, areas remain for further enhancement. For example, integrating transformer-based architectures, which have shown great promise in vision tasks, could further enhance the model's ability to capture long-range image dependencies. Additionally, the model could be extended to handle more complex functions like instance segmentation or object tracking. These future advancements will likely continue to push the boundaries of what is possible in real-time object detection.

2) BACKGROUND SUBTRACTION AND FOREGROUND EXTRACTION

Background subtraction is a widely used technique in computer vision for separating moving objects (foreground) from the static environment (background) in a video sequence. This process is fundamental in detecting and tracking video objects, especially in applications such as surveillance, traffic monitoring, and human activity recognition [36]. The method operates by modelling the scene's background and detecting changes corresponding to moving objects, considered the foreground. The efficiency and accuracy of background subtraction depend on the robustness of the model used to handle dynamic changes in the background, such as lighting variations, shadows, or repetitive motions like tree branches swaying in the wind. One of the most common approaches

to background subtraction is the *Gaussian Mixture Model (GMM)* [37], where each pixel's intensity is modelled as a mixture of Gaussians. This model captures the statistical distribution of the pixel values over time and classifies each pixel as background or foreground based on its probability distribution. Mathematically, for each pixel value x_t at time t , the GMM can be expressed in Equation 5:

$$P(x_t) = \sum_{i=1}^K w_i \mathcal{N}(x_t | \mu_i, \Sigma_i) \quad (5)$$

where w_i is the weight, μ_i is the mean, and Σ_i is the covariance of the i -th Gaussian component. If the current pixel value deviates significantly from all Gaussian distributions in the model, it is classified as part of the foreground.

Foreground extraction is the next crucial step after background subtraction. It isolates the detected moving objects from the rest of the frame, which can be processed further for tasks like tracking or recognition. The extracted foreground is often represented as a binary mask, where the pixels belonging to moving objects are marked as foreground (white) and the rest as background (black). In practical implementations, morphological operations like erosion and dilation are applied to this mask to remove noise and refine object boundaries, ensuring a cleaner and more accurate foreground. Another technique for background subtraction is the *Running Average Method*, which updates the background model dynamically over time by averaging the pixel intensities [38]. The background at time t , $B_t(x, y)$, is updated based on Equation 6:

$$B_t(x, y) = (1 - \alpha)B_{t-1}(x, y) + \alpha I_t(x, y) \quad (6)$$

where $I_t(x, y)$ is the current pixel intensity, and α is the learning rate that controls how quickly the model adapts to changes. This method is simple and effective for static backgrounds but may struggle in dynamic environments.

Foreground extraction often involves post-processing steps to refine the result. Techniques like *morphological operations* (erosion and dilation) can remove slight noise and close gaps in the detected foreground [39]. *Erosion* shrinks the foreground by removing isolated pixels, while *dilation* expands the foreground to recover lost information [40]. These operations help produce a cleaner mask for subsequent object detection or tracking tasks.

In addition to traditional methods, deep learning has significantly improved background subtraction and foreground extraction. *Convolutional Neural Networks (CNNs)* have been used to learn complex features from video data, enabling the model to distinguish between background and foreground with higher accuracy [5]. CNN-based models can automatically learn robust features that account for lighting changes, camera jitter, and other challenges that traditional methods struggle with. For example, a CNN can be trained on labelled data where the background and foreground have been manually segmented, allowing the model to generalize well to unseen video sequences. For scenarios where the

background is not entirely static, such as outdoor scenes with moving vegetation, *Optical Flow* can be used in conjunction with background subtraction [41]. Optical flow computes the motion between consecutive frames, estimating the velocity of objects in the scene. The optical flow field is then used to distinguish between static background elements and moving foreground objects. Mathematically, the optical flow vector $\mathbf{v} = (u, v)$ represents the motion of each pixel, where u and v are the horizontal and vertical displacements, respectively. The magnitude of this vector helps determine whether a pixel belongs to the foreground or background. To deal with challenges like lighting changes, *Illumination Invariance* techniques can be incorporated into background subtraction algorithms [42]. These methods rely on features invariant to changes in lighting conditions, such as gradient information, instead of raw pixel intensities. Using gradient-based features, background subtraction algorithms can better handle scenes where lighting conditions vary significantly, ensuring that foreground objects are detected accurately.

Foreground extraction from the background subtraction mask also benefits from advanced techniques such as *Region Growing* and *Contour Detection* [43]. Region-growing methods expand detected regions by grouping neighbouring pixels with similar properties (e.g., colour or intensity), while contour detection identifies the boundaries of moving objects. These techniques are instrumental when precise object boundaries are needed for further tasks like object classification or tracking. The success of background subtraction and foreground extraction in detecting objects from videos depends heavily on how well the system can adapt to dynamic scenes. This involves handling challenges such as moving shadows, reflections, and camera motion. Advanced methods incorporate *Shadow Removal* techniques to prevent shadows cast by moving objects from being incorrectly classified as part of the foreground [44]. This is often achieved by analyzing color and intensity differences between the detected foreground object and the background model. Finally, integrating background subtraction with *Deep Learning Models* like *YOLOv8* for object detection has shown significant promise. Background subtraction helps reduce the computational load by isolating the region of interest, allowing the object detection model to focus only on the foreground. This hybrid approach combines the efficiency of traditional background subtraction methods with the accuracy of deep learning models, leading to more effective and real-time object detection in complex video scenes.

3) TIME-SPACE TRANSFORMER (TST)

The *TST* is an advanced model for handling spatial and temporal data in video object detection tasks [45]. This model extends the traditional transformer architecture, initially designed for natural language processing, by adapting it for spatio-temporal tasks in computer vision. The TST processes video sequences by modelling objects' spatial features within individual frames and the temporal dynamics between

consecutive frames. This dual capability makes it particularly useful for detecting and tracking objects in videos, where space and time play crucial roles. In object detection using video, each video sequence frame is treated as a spatial data input, while the sequence of frames forms the temporal data. The TST encodes these two aspects by dividing the input into patches, similar to the Vision Transformer (ViT) [46]. Each video frame is split into a grid of patches, and each patch is treated as a token in the transformer. The spatial relationships between these patches within a single frame are modelled using self-attention mechanisms. In contrast, the temporal relationships between patches from different frames are handled by extending the self-attention across time. Mathematically, the input sequence $X \in \mathbb{R}^{T \times H \times W \times C}$, where T represents the number of frames, $H \times W$ is the spatial dimension of each frame, and C is the number of channels, is reshaped into a sequence of patches and processed by the transformer.

The core of the transformer architecture is the *self-attention mechanism*, which computes the importance of each token (patch) relative to all other tokens in the sequence [47]. For the Time-Space Transformer, the self-attention operates across both spatial and temporal dimensions. Given a query Q , key K , and value V matrices, the attention score for each patch is computed using Equation 7:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (7)$$

where d_k is the dimensionality of the key vectors, and the softmax function ensures that the attention scores are normalized. In the TST, self-attention is applied within the patches of a single frame (spatial attention) and between corresponding patches across multiple frames (temporal attention). This allows the model to capture spatial dependencies (how objects are arranged in a single frame) and temporal dependencies (how objects move across frames).

One of the challenges in applying transformers to video data is ensuring that the model accounts for the temporal order of the frames. In natural language processing, positional encodings are added to the input tokens to retain the order of words in a sentence. Similarly, in the Time-Space Transformer, *temporal positional encodings* is added to the patch embeddings to encode the order of the video frames [48]. The temporal encoding for each frame at time t is defined in Equation 8:

$$\text{PE}_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad \text{PE}_{t,2i+1} = \cos\left(\frac{t}{10000^{2i/d}}\right) \quad (8)$$

where d is the dimensionality of the embedding, and i is the index of the positional encoding. These encodings allow the transformer to learn the temporal relationships between frames, ensuring that the model can track the motion of objects over time.

To further enhance the model's ability to detect objects in videos, the *multi-head attention* mechanism is employed [49].

In this approach, the self-attention is computed multiple times (in different "heads"), each focusing on various aspects of the input data. Each head has its query, key, and value matrices, and the results from all heads are concatenated and linearly transformed. The multi-head attention mechanism can be described in Equation 9:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W_O \quad (9)$$

where each head is computed as:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (10)$$

And W_O is the output weight matrix. This allows the Time-Space Transformer to simultaneously focus on different parts of the scene and capture global and local dependencies, improving the detection of objects in complex video scenarios.

Another critical feature of the Time-Space Transformer is the use of *cross-attention* layers. Cross-attention allows the model to fuse information from multiple sources, such as spatial and temporal features [50]. This is particularly useful in object detection because it enables the model to combine the spatial features extracted from individual frames with the temporal features representing object motion. Cross-attention operates by computing the attention between the query from one modality (e.g., spatial features) and the key-value pairs from another (e.g., temporal features). This fusion improves the model's ability to detect objects that may not be easily distinguishable in a single frame but become more apparent when considering their motion across multiple frames. In the Time-Space Transformer, each video frame is divided into patches, and each patch is embedded into a high-dimensional space before being passed through the transformer layers. This embedding step is crucial for encoding both spatial and temporal information. The embedding for each patch can be represented in Equation 11:

$$\mathbf{z}_p = W_p \mathbf{x}_p + \mathbf{e}_p \quad (11)$$

where $\mathbf{x}_p$ is the raw pixel data for patch p , W_p is a learned weight matrix, and $\mathbf{e}_p$ is the positional encoding added to retain spatial or temporal information. The transformer then operates on these embeddings, allowing the model to learn complex spatiotemporal patterns critical for video object detection.

One of the challenges in detecting objects in video sequences is dealing with noisy or erratic motion. To address this, the Time-Space Transformer incorporates *temporal smoothing* techniques that regularize the attention weights across frames [51]. This ensures that the model pays consistent attention to the same object over time, even if the object undergoes minor appearance changes due to occlusion or lighting variations. Temporal smoothing is achieved by adding a regularization term to the loss function, which penalizes significant variations in the attention weights

between consecutive frames, as expressed in Equation 12:

$$L_{\text{smooth}} = \sum_{t=2}^T \|\mathbf{A}_t - \mathbf{A}_{t-1}\|^2 \quad (12)$$

where $\mathbf{A}_t$ is the attention weight matrix at time t , and the loss encourages smooth transitions between frames.

Objects can appear in complex video scenes at different scales, making detection challenging. The Time-Space Transformer addresses this by incorporating a *multi-scale attention mechanism*, where attention is computed at multiple spatial resolutions [52]. This allows the model to detect small objects in fine-grained detail while capturing larger ones at a coarser scale. The multi-scale feature maps are fused to produce a unified representation that captures objects of all sizes, improving the model's overall performance in detecting objects at varying scales. Despite the complexity of the Time-Space Transformer, it can be optimized for real-time applications by leveraging techniques such as *knowledge distillation* and *model pruning* [53]. Knowledge distillation involves training a smaller, faster model (student) to mimic the behaviour of a larger, more complex model (teacher). By transferring knowledge from the teacher to the student model, the Time-Space Transformer can be lightweight enough for real-time object detection tasks, such as surveillance or autonomous driving.

B. RESEARCH SIGNIFICANCE

The motivation behind this research stems from the growing need for robust and reliable fall detection systems, particularly for vulnerable populations such as the elderly. Falls represent a significant health risk, often leading to serious injuries, extended hospital stays, and even fatalities if not addressed promptly. Therefore, early detection of falls is crucial in minimizing injury severity and enabling timely medical intervention. This research focuses on creating a system that offers real-time fall detection using video-based techniques that are less intrusive than wearable sensors, providing a more comfortable and scalable solution for monitoring falls in various environments like homes, hospitals, and nursing facilities. Traditional fall detection methods have typically relied on wearable sensors, which are prone to several limitations, including user discomfort, maintenance challenges, and reduced accuracy due to improper placement or battery depletion. These challenges highlight the need for non-intrusive, video-based methods that can function autonomously without requiring the user to wear any device. The significance of this study lies in addressing these shortcomings by leveraging video data to enhance detection capabilities while maintaining the ease of deployment and user comfort. When combined with deep learning, video-based detection systems offer a powerful alternative by focusing solely on environmental cues to detect falls.

In addition to providing comfort and ease of use, video-based fall detection systems must be accurate and efficient. One of the primary challenges in this area is achieving high

accuracy while minimizing false positives and negatives. Falls can be subtle and vary widely in how they manifest, making it difficult for conventional models to distinguish them from everyday activities like sitting down quickly or bending over. To tackle these challenges, this research integrates state-of-the-art deep learning models, specifically YOLOv8 for object detection and Time-Space Transformers for temporal analysis. The hybrid approach enhances spatial and temporal awareness, improving fall detection accuracy significantly. The novel contribution of this research is the synergy between YOLOv8 and Time-Space Transformers. YOLOv8 detects humans in individual video frames with high accuracy and speed. At the same time, Time-Space Transformers capture the temporal dynamics of human movement, making it possible to detect abnormal postural changes that indicate a fall. The combination of these two models allows for real-time, non-intrusive monitoring, providing a more holistic and reliable solution to the problem of fall detection. The significance of this hybrid approach is its potential to outperform single-model systems, which may struggle to capture both spatial and temporal complexities.

Furthermore, the real-time processing capabilities of the proposed system are vital for practical applications. A fall detection system that fails to respond quickly may result in delayed assistance, negating the purpose of early intervention. YOLOv8's efficiency in object detection ensures that the system can operate in real-time, while Time-Space Transformers allow for swift analysis of motion patterns across frames. This provides the detection accuracy and the system's ability to alert caregivers or medical personnel promptly, reducing response time and potentially saving lives. Finally, this research is significant for real-world deployment, especially in an ageing global population. As the number of elderly individuals continues to rise, the demand for efficient, scalable, and user-friendly fall detection systems will only increase. By integrating cutting-edge deep learning models, this research paves the way for more robust and autonomous monitoring solutions that can be deployed in private homes and healthcare institutions, offering peace of mind to caregivers and enhancing the safety and well-being of individuals at risk of falls.

C. PROBLEM STATEMENT AND LIMITATIONS OF CURRENT METHODS

Current fall detection methods face several limitations that hinder their effectiveness in real-world applications. While widely used, wearable sensor-based approaches are often intrusive and uncomfortable for users, leading to low adoption rates, particularly among elderly individuals. These devices require frequent maintenance, including battery charging and proper placement, which impacts their long-term usability. Also, false alarms are common due to normal human activities resembling falls, such as sudden sitting or bending movements. These inaccuracies reduce trust in the system and may lead to alarm fatigue, where caregivers disregard alerts due to frequent false positives. Furthermore,

wearable sensors may fail to function effectively in emergencies, as individuals experiencing a fall may be unable to interact with the device to trigger an alert.

On the other hand, video-based fall detection systems, while offering a non-intrusive alternative, face their challenges. Many traditional methods rely on handcrafted features or shallow machine learning models, which struggle to generalize across diverse environments. Variations in lighting, camera angles, background clutter, and occlusions often lead to poor performance in real-world scenarios. Additionally, distinguishing subtle differences between actual falls and normal human activities (e.g., sitting down quickly, tying shoelaces) remains a major challenge, leading to high false positive rates. Furthermore, conventional video-based approaches often operate on frame-by-frame analysis, failing to capture temporal dependencies essential for recognizing fall patterns accurately.

Some methods address this by incorporating sequential models such as LSTMs and GRUs. Still, these models suffer from vanishing gradient issues and are computationally expensive, making them unsuitable for real-time applications. Lastly, privacy concerns remain a significant barrier to adoption, as many video-based systems require continuous surveillance, raising ethical and regulatory challenges in sensitive environments such as homes and healthcare facilities.

D. RESEARCH OBJECTIVES

The proposed hybrid deep learning approach integrates YOLOv8 for real-time human detection and TST for temporal motion analysis to address the previously mentioned gaps and limitations. This framework aims to enhance spatial and temporal awareness in fall detection, ensuring high accuracy while reducing false positives and negatives. The YOLOv8 component effectively detects and localizes humans within video frames, allowing for precise bounding box tracking. Meanwhile, the TST model captures long-range temporal dependencies, distinguishing abrupt falls from normal daily activities by learning motion patterns across consecutive frames. This hybrid approach improves fall detection performance and ensures computational efficiency, making it suitable for real-time deployment in healthcare and smart home environments. Additionally, by leveraging self-attention mechanisms, the system overcomes the limitations of sequential models like LSTM and GRU, which struggle with long-range dependencies. The proposed system is designed to offer a scalable, non-intrusive, and high-performance fall detection solution, ensuring reliable monitoring and timely interventions for at-risk individuals.

Thus, this paper aims to develop an efficient, non-intrusive, real-time fall detection system by integrating advanced deep learning models, specifically YOLOv8 and Time-Space Transformers, to enhance spatial and temporal awareness for accurate fall detection in real-world environments. The objectives of the paper are as follows:

- Develop a non-intrusive, real-time fall detection system that leverages image-based methods and deep learning models for accurate monitoring without needing wearable devices.
- Enhance detection accuracy and minimize false positives/negatives by integrating YOLOv8 for spatial object detection and Time-Space Transformers for temporal motion analysis, allowing the system to differentiate between falls and normal activities.
- Design a scalable and reliable fall detection system suitable for deployment in various real-world environments such as homes and healthcare facilities, ensuring timely alerts and ease of use for caregivers and vulnerable populations.

The subsequent sections of this paper are organized as follows: *Section II: Literature Review* discusses existing work in the area of fall detection and image-based techniques, along with a review of deep learning models used in object detection and motion analysis. *Section III: Methodology* describes the proposed system in detail, outlining the steps involved in image acquisition, preprocessing, human detection using YOLOv8, temporal analysis with Time-Space Transformers, and the final fall detection decision-making process. *Section IV* presents the experiments setups in details. *Section V: Results and Discussion* presents the experimental setup, evaluation metrics, and the results obtained from applying the proposed model. The performance of the hybrid model is analyzed and compared with other approaches. Limitations and future works are presented in *Section VI*. Finally, *Section VII: Conclusion* summarizes the paper's contributions and conclusions.

II. LITERATURE REVIEW

This section provides an overview of the research advancements in human fall detection systems, particularly emphasizing video-based approaches enhanced by deep learning models. With the growing need to ensure the safety of vulnerable populations, such as the elderly, significant research efforts have focused on developing systems that can detect falls accurately and efficiently. Although widely used, traditional sensor-based methods have limitations regarding comfort and maintenance. As a result, video-based fall detection systems have emerged as a more scalable and non-intrusive alternative. Recent works have leveraged powerful deep learning models like YOLOv8 and Time-Space Transformers to improve the detection of falls in real-time by analyzing both spatial and temporal dynamics. This review aims to synthesize current methodologies, challenges, and future directions in fall detection, highlighting the effectiveness of hybrid deep learning systems in enhancing detection accuracy while minimizing false positives and negatives.

Recent advancements in deep learning have significantly enhanced real-time detection and medical imaging applications, with a growing emphasis on model explainability, computational efficiency, and accuracy. The integration of

explainable AI (XAI) techniques has been particularly instrumental in improving the interpretability of deep learning models in critical domains such as medical diagnostics and human activity recognition.

Hussain et al. [54] explored the role of XAI in deep learning-based brain tumor classification, demonstrating how techniques like Grad-CAM and Grad-CAM++ can provide visual explanations to enhance decision transparency. Their findings highlight the necessity of incorporating interpretability mechanisms in AI-driven systems, particularly in applications where human trust and reliability are paramount. Similarly, Alam et al. [55] leveraged YOLOv8 alongside deep feature extraction models to refine region-based detection for fracture classification, illustrating the effectiveness of transformer-based architectures and CNNs in high-precision object detection tasks. Furthermore, Hussain et al. [56] proposed a multi-attention gated residual U-Net (MAGRes-UNet) for medical image segmentation, addressing challenges related to feature extraction, model complexity, and small-scale object recognition. Their work underscores the importance of attention mechanisms and residual learning in refining deep learning models for improved robustness and accuracy.

Khan et al. [57] addressed the critical need for real-time intelligent fall detection systems, particularly for elderly individuals, patients with neurological or cardiovascular diseases, and disabled children. While multimodal approaches are commonly used, they are often computationally expensive due to the integration of multiple sensors. Computer vision-based systems offer an alternative but struggle with real-time performance and visual intelligence integration at feasible stages. Additionally, the lack of large-scale, well-annotated datasets limits fall detection accuracy in complex environments. To overcome these problems, the study proposes an enhanced version of YOLOv8 for fall detection. Key contributions include the introduction of a large-scale dataset comprising approximately 10,500 annotated images designed to improve model generalization in diverse environments. The study also presents progressive improvements to the YOLOv8S model, incorporating a focus module in the backbone for optimized feature extraction. Furthermore, convolutional block attention modules are added at specific stages to enhance spatial and channel context understanding, which is crucial for accurate detection in complex scenes. Extensive empirical evaluations demonstrate the proposed model's superiority over 13 state-of-the-art techniques, supported by benchmarking and qualitative validation across various environments. The results show that the enhanced YOLOv8 model excels in performance, size, and processing time, making it an effective solution for real-time fall detection.

Chouhan et al. [58] proposed an automated human fall detection system utilizing deep learning techniques to enhance accuracy and efficiency. The model uses a dataset containing falling and non-falling images processed through

a CNN. The CNN is employed for its ability to recognize patterns and features in the photos, facilitating effective classification of fall-related incidents. Several limitations were addressed, such as the lack of large-scale datasets and human errors by ensuring improved generalization and robustness in diverse environments. Furthermore, the system emphasizes the security and privacy of sensitive information contained in the images. Experimental results on real-world fall datasets show that the CNN model achieves higher accuracy and reduced loss, making it a promising solution for real-time fall detection and prevention.

Proposed a sensor-based alarm system [59] to detect human falls. The proposed system was introduced to assist elderly individuals prone to falling due to various health conditions. The proposed system utilizes an architecture that differentiates fall events from other indoor human activities. The method begins with a video frame generator that extracts frames from video clips, followed by feature extraction using a two-dimensional CNN (2DCNN). A gated recurrent unit network is then employed to capture the temporal dependencies of human movements. The network's performance is optimized using a binary cross-entropy loss function, which adjusts parameters like weights and learning rates to reduce losses. For the final stage, a sigmoid classifier is used to identify fall events through binary classification. Experimental results demonstrate that the model achieves an impressive accuracy of 99%, outperforming other contemporary models in fall detection. Another study proposed by [60] addresses the problem of detecting falls in indoor environments, particularly for elderly and ill individuals who live alone. The proposed solution is a deep learning-based framework for fall detection and classification using continuous camera monitoring, which minimizes user interaction. In addition to deep learning, traditional machine learning methods such as Support Vector Machine (SVM) and decision tree classifiers are applied after feature extraction to detect and classify falls. The study compares the deep learning framework with SVM, decision trees, and other state-of-the-art approaches using two publicly available datasets. The experimental results indicate that the proposed deep learning framework delivers promising performance and is comparable to other advanced fall detection methods. In [61], the authors focus on the early detection of high fall risks in elderly individuals using wearable sensors, such as gyroscopes and accelerometers, which effectively monitor daily activities. The research implements deep learning methods to analyze sensor data and assess fall risks. A publicly available dataset of elderly individuals' daily activities is utilized for the analysis. The study compares the performance of two deep learning architectures: Long Short-Term Memory (LSTM) and CNN-based Transfer Learning. The results indicate that CNN-based Transfer Learning achieves superior performance, with 98% accuracy. The findings highlight that deep learning models, particularly in the context of wearable sensors, are highly effective in predicting fall risks and supporting multi-task learning.

In [62], the authors focus on the growing need for advanced fall detection systems to assist in elderly care, especially given the global ageing population. Traditional manual monitoring methods have shifted towards virtual systems that provide more accurate, real-time monitoring. The authors propose a model for real-time fall detection utilizing deep CNN, highlighting the importance of automated systems in preventing fatalities from falls by offering immediate emergency assistance. The proposed system leverages deep learning techniques and torch tensors to distinguish between fall and non-fall events in real-time video input. A hardware demonstration was conducted using an Up Extreme Series single-board computer and an LSTM neural network model, showcasing the system's ability to recognize falls and respond promptly. This approach emphasizes the efficiency and necessity of real-time fall detection systems in providing timely care to the elderly.

The authors of [63] proposed a fall detection system designed for smart home environments, leveraging technologies such as the Internet of Things (IoT) and artificial intelligence (AI). The proposed system uses cameras to monitor activities and detect falls, addressing the discomfort associated with wearable devices. The fall detection model is based on the deep CNN Inception v3 architecture (CNN-Inception v3) and is designed to integrate smart homecare systems. The system is evaluated using multiple camera fall datasets and the UR fall dataset, demonstrating its effectiveness in identifying falls in real-time.

The unnoticed falls among elderly individuals living alone problem was addressed by [64] for life-saving treatment during the 'golden hour.' To address this problem, the authors have developed a wrist-wearable fall detection system that utilizes machine learning for movement tracking and fall detection. Upon detecting a fall, the system automatically triggers an emergency call to both emergency services and a designated user via a GSM module. Two datasets were collected for this study, and seven different machine learning models were trained and tested, with the results presented in the paper. Also, in [65], the authors handle the growing concern of falls among elderly individuals, particularly those living alone, which can lead to serious health risks if not immediately attended to. Falls often result from age-related mobility issues such as reduced fitness, muscle strength, and balance. Current fall detection technologies are classified into two main types: wearable sensors and ambient sensors. Wearable devices, like pendants and wristbands, use accelerometers to detect falls and often include manual alert buttons, requiring the user's familiarity with the technology. Ambient sensors like video cameras monitor movements and trigger alerts based on fall events. The authors propose a video-based fall detection system using security cameras, computer vision, and deep learning algorithms to accurately distinguish falls from regular activities. The system includes multiple-angle video analysis and multi-person tracking to enhance detection accuracy and estimate the severity of falls,

allowing for prompt emergency responses. This system aims to improve safety, independence, and timely assistance for individuals at risk of falling.

Another research addressed the same problem [66] in an IoT environment. The proposed solution employs a deep learning-based fall detection system using a CNN with three hidden layers. The model was evaluated using a fall detection dataset, achieving a classification accuracy of 96.5%. However, while deep learning models are effective, their "black-box" nature limits trust among healthcare professionals. To address this, the paper introduces an Explainable AI model called LIME (Local Interpretable Model-Agnostic Explanations), which interprets the classification process by highlighting the features responsible for predicting fall activities. The LIME results enhance the transparency of the model, making it more acceptable for healthcare applications.

Chen et al. [67] introduced a Wi-Fi sensing-based approach as a privacy-preserving method for detecting falls indoors. Wi-Fi signals, reflected off objects and individuals within the environment, carry valuable information about their orientation and activity. These reflected signals are captured by receivers, which aggregate data to detect human activities, including falls. The authors collected a dataset in practical experimental environments and applied deep learning techniques to classify fall events. This method offers a practical, non-invasive solution for monitoring the elderly while ensuring privacy. Further, [4] enhanced the fall detection solutions in smart home environments. While various vision and non-vision-based techniques exist, vision-based methods have garnered significant attention due to their feasibility and applicability. However, vision-based fall detection systems often suffer from low accuracy and high computational costs. To address these limitations, this research introduces a new dataset for posture and fall detection, with images sourced from the internet and enhanced through data augmentation. The study employs different YOLO networks for fall detection, comparing YOLOv5n, YOLOv5s, and YOLOv6s models based on accuracy, F1 score, recall, and mAP. Experimental results indicate that YOLOv5s outperformed the other versions in terms of accuracy and overall effectiveness. Finally, the authors of [2] have reviewed the critical issue of human fall detection, particularly for elderly and disabled individuals living alone. The paper highlights the increasing use of deep learning and computer vision techniques in non-intrusive, vision-based fall detection systems designed to support assistive living. The authors present a comprehensive survey of state-of-the-art deep learning methods and benchmark datasets used in fall detection research. Additionally, they discuss various evaluation metrics employed to measure the performance of these systems.

Table 1 summarises key studies on human fall detection, highlighting their objectives, methodologies, findings, and notable weaknesses. The existing research has made significant progress in fall detection through various approaches,

including deep learning, sensor-based systems, and video-based methods. However, several gaps remain, such as the reliance on static datasets, limited real-time performance, high computational costs, and poor generalizability across different environments. These limitations create challenges in deploying reliable, scalable, and efficient fall detection systems, especially in real-world settings where diverse scenarios and rapid detection are crucial. Thus, the proposed human fall detection model aims to address these gaps by focusing on three main objectives: (1) developing a real-time, non-intrusive system that leverages video-based methods and hybrid deep learning models (YOLOv8 and Time-Space Transformers) for accurate fall detection, (2) enhancing both spatial and temporal awareness to minimize false positives and negatives, particularly in dynamic environments, and (3) creating a system that is computationally efficient and scalable for real-world applications, including smart homes and healthcare settings. The contributions of this paper lie in optimizing the model for real-time performance while maintaining high detection accuracy, improving generalizability across diverse conditions, and making it suitable for resource-constrained environments. These improvements directly map to the identified gaps in the literature, ensuring that the proposed model offers a significant advancement over previous approaches.

III. METHODOLOGY

This section outlines the approach to developing the proposed fall detection system, which integrates YOLOv8 for real-time human detection and Time-Space Transformers for temporal motion analysis.

Figure 1 illustrates six distinct types of falls, each representing a unique challenge for a human fall detection system. These fall types—stumbling, slipping, fainting, falling from a sitting position, falling from a height, and jumping down followed by a fall—provide a comprehensive set of scenarios that a fall detection system must recognize and categorize accurately to ensure a timely and effective response. In the case of *stumbling*, the fall detection system must recognize sudden, irregular shifts in balance where the person may not necessarily make ground contact. Stumbling involves an abrupt loss of balance, which might only be temporary and could be mistaken for normal movement by less sophisticated systems. A detection system must account for such erratic motion and differentiate it from actual falls. The challenge here is to ensure that the system doesn't trigger false positives by interpreting a minor loss of balance as a whole fall. *Slipping*, introduces more pronounced motion patterns, characterized by a sudden loss of traction leading to a rapid descent, usually backward or forward. In such cases, the system must recognize both the initial imbalance and the subsequent impact, typically involving sharp accelerations and decelerations. The quick and unplanned movement associated with slipping requires the system to track rapid changes in motion and acceleration. A reliable fall detection system should be sensitive enough to detect these patterns in

real-time, ensuring that it can differentiate between slips and normal walking or running movements.

The next type, *fainting*, poses different challenges. Unlike slips and stumbles, fainting usually leads to a slower, more controlled collapse as the body loses consciousness or muscle control. The fall detection system needs to be adept at identifying a gradual loss of balance and slower descent. This type of fall may not exhibit rapid movement in other scenarios. Still, its detection is critical, as fainting incidents often result in injuries or dangerous situations, mainly if the person is alone. Systems must capture the distinct motion signature of fainting, recognizing the lack of corrective movement and the eventual ground contact. In *falling from a sitting position*, such as from a chair, the system must differentiate between the natural, slower transitions during seated activities and actual falls. While the fall distance might be shorter, the risk of injury is still present, particularly for the elderly or those with mobility impairments. The system should be tuned to recognize abnormal accelerations and body movements that indicate an involuntary fall from a seated posture. This type of fall detection requires balancing sensitivity with the ability to ignore regular sitting movements.

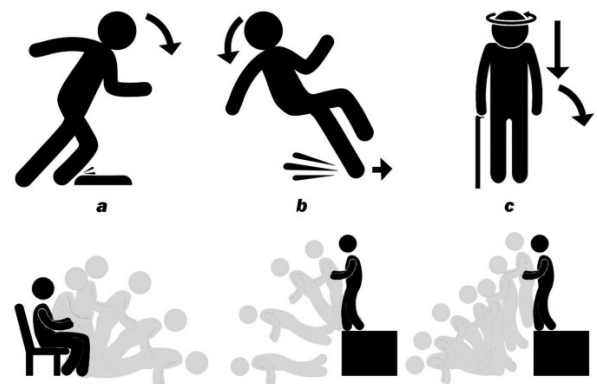


FIGURE 1. Fall types.

Falling from a height, such as stairs or ladders, introduces high-velocity movements that the system must track effectively. These falls typically occur over a longer time frame, as the person falls from a higher point, potentially leading to severe injuries. The system must quickly recognize the elevated height and velocity of the fall and trigger an alert well before the impact occurs. High-speed falls, combined with more extended time in motion, make it essential for the fall detection system to capture both the trajectory and the time frame of the descent to avoid delayed responses.

Finally, *jumping down followed by a fall* involves voluntary motion (the jump) followed by uncontrolled impact (the fall). This scenario presents a unique challenge, as the system must differentiate between voluntary jumps, which may include fast movement, and actual falls, which follow a loss of control upon landing. The fall detection system must capture the moment of loss of power and subsequent impact, ensuring

TABLE 1. Summary of related works and evaluation.

Study	Objective	Method(s)	Findings	Weaknesses	Contributions of the Proposed Solution
[57]	Develop a real-time fall detection system using YOLOv8 for elderly care	Enhanced YOLOv8	Achieved superior performance over 13 models.	Dataset lacks diversity in environmental conditions, limiting generalization.	The proposed solution builds on this by integrating TST to enhance temporal awareness, improving performance across diverse conditions.
[58]	Create an automated fall detection system	CNN	High accuracy and reduced loss.	Limited to static image data lacks temporal analysis for real-time scenarios.	The proposed solution incorporates temporal modelling via TST, ensuring effective real-time fall detection by capturing motion sequences.
[59]	Design a sensor-based alarm system for fall detection	2DCNN and GRU	Achieved 99% accuracy.	High computational cost, unsuitable for resource-constrained environments.	We optimize computational efficiency by using a hybrid deep learning approach, balancing accuracy with real-time performance.
[60]	Compare fall detection methods in indoor settings	Deep learning and traditional machine learning models	Comparable results to state-of-the-art methods.	Complexity from multi-sensor integration increases computational overhead.	The proposed solution eliminates the need for multi-sensor integration by leveraging vision-based approaches, making it more efficient and scalable.
[61]	Predict fall risks	LSTM and CNN	CNN-Transfer Learning achieved 98% accuracy.	Focuses solely on wearable sensor data, limiting applicability to video-based solutions.	We address this limitation by introducing a fully vision-based approach, eliminating the reliance on wearables while maintaining high accuracy.
[62]	Real-time fall detection	Deep learning with hardware integration	Effective fall recognition in real-time.	Hardware-specific approach limits flexibility and adoption in diverse environments.	We introduce a hardware-independent, deep learning-based model adaptable to multiple deployment settings.
[63]	Develop fall detection using IoT and AI in smart homes	CNN-Inception v3	Effective fall detection in real-time.	Evaluations constrained to specific datasets, limiting generalizability.	The proposed solution is trained on multiple diverse datasets, ensuring broader generalization and applicability in real-world scenarios.
[64]	Create a wrist-wearable fall detection system	Machine learning to track movement	Successfully triggers emergency calls.	Dependent on wearable technology with limited dataset, affecting scalability.	The proposed solution eliminates the need for wearables, making it scalable and deployable in a wide range of environments.
[65]	Develop a video-based fall detection system	Computer vision and deep learning	Effective multi-angle video analysis for fall detection.	High computational cost, limited optimization for real-time detection.	We optimize deep learning models for real-time efficiency, reducing computational overhead while maintaining accuracy.
[66]	Enhance fall detection with explainable AI	CNN with LIME	Achieved 96.5% accuracy with explainable insights.	Lacks comprehensive explanation of decision-making, reducing trust.	The proposed solution enhances interpretability by integrating explainable AI techniques alongside robust fall detection.
[67]	Use Wi-Fi sensing for fall detection in smart homes	Deep learning	Effective privacy-preserving fall detection.	Limited to indoor environments with Wi-Fi coverage.	The proposed solution removes the dependency on Wi-Fi-based sensing, making it adaptable to outdoor and varied indoor settings.
[4]	Improve fall detection accuracy using YOLO models	YOLOv5n, YOLOv5s, and YOLOv6s	YOLOv5s had the best accuracy and performance.	High computational cost, challenging for real-time deployment.	We integrate YOLOv8 with TST, optimizing performance while reducing computational costs for real-time applications.
[2]	Review deep learning and computer vision techniques for fall detection	Survey of state-of-the-art methods and benchmark datasets	Provided a comprehensive overview of fall detection technologies.	Lacked practical implementation details for real-world scenarios.	The proposed solution extends this by implementing and testing a hybrid model on real-world datasets, validating practical deployment feasibility.

that jumps are not mistaken for falls unless the landing results in an unintended collapse.

In sum, these different types of falls require a human fall detection system to handle a variety of motion patterns and impacts. The system must balance sensitivity with the ability to filter out non-fall activities, ensuring high accuracy and minimal false positives. Advanced machine learning models, such as those integrating YOLOv8 and time-space transformers, are well-suited for this task, as they can process real-time video data and recognize spatial and temporal dependencies in human movements. By learning the distinctive motion patterns of each type of fall, these models can improve fall detection, ensuring faster and more accurate alerts for caregivers or emergency personnel.

As depicted in Figure 2, the methodology addresses both spatial and temporal challenges in accurately detecting

human falls from video sequences (i.e., images). The process begins with image input acquisition, followed by background subtraction and foreground extraction to isolate relevant objects. YOLOv8 is then employed to detect humans in each frame, while the Time-Space Transformer analyzes motion across consecutive frames to identify abnormal patterns indicative of a fall. The steps involved, including preprocessing, feature extraction, and decision-making for fall detection, are detailed to provide a comprehensive understanding of the system's architecture and functionality.

Step 1 (Image Input Acquisition): The initial step involves utilizing a benchmark dataset called CAUCAFall, which consists of video recordings of human subjects performing various activities, including falls. The dataset is structured to include both normal activities (sitting, walking) and fall events, allowing for comprehensive model training and

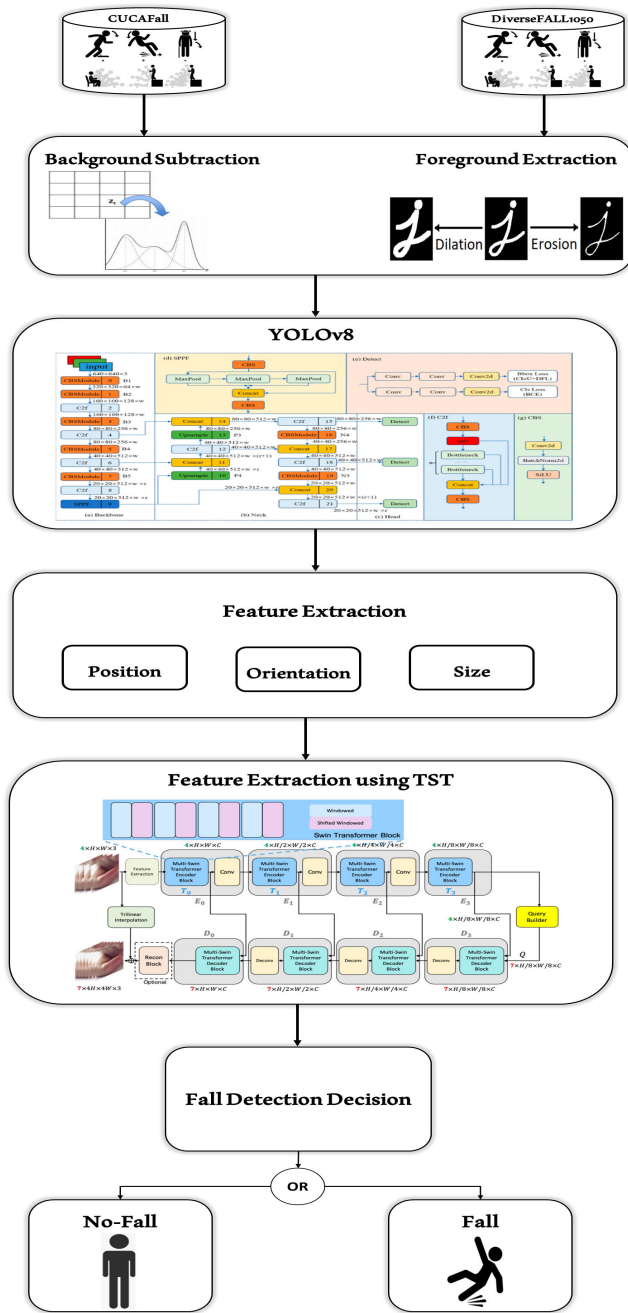


FIGURE 2. Proposed methodology.

evaluation. The videos are captured from multiple angles and in varying environments, ensuring the model can generalize to diverse real-world conditions. Each video in the dataset is pre-labelled with ground truth annotations, identifying whether a fall occurred and, if so, the precise frames in which it happened. Also, the DiverseFALL10500 dataset is a large-scale benchmark designed for fall detection research, containing around 10,500 samples. Each sample is annotated to differentiate between fall and non-fall instances, offering a valuable resource for training and evaluating detection models. It includes diverse environmental conditions and

scenarios, enhancing the model's generalization ability and making it suitable for real-time applications.

Step 2 (Preprocessing and Background Subtraction): Preprocessing is essential for ensuring high-quality input data by preparing the video sequences for analysis. This involves multiple techniques, including background subtraction, foreground extraction, data augmentation, normalization, and noise reduction, to enhance model robustness and improve fall detection accuracy.

- **Background Subtraction:** A background subtraction algorithm, such as the Gaussian Mixture Model (GMM), separates the foreground (moving human) from the static background. Each pixel's intensity in a frame is modelled as a mixture of several Gaussians representing the background. The probability of a pixel x_t at time t belonging to the background is given by:

$$P(x_t) = \sum_{i=1}^K w_i N(x_t | \mu_i, \Sigma_i) \quad (13)$$

where w_i , μ_i , and Σ_i are the weight, mean, and covariance of the i -th Gaussian component. The GMM model is updated dynamically to account for background variations over time. This method ensures that only moving objects like humans remain in the extracted foreground.

- **Foreground Extraction:** Once the background has been subtracted, the foreground, which typically contains the human subject, is extracted as a binary mask. Morphological operations such as *erosion* (to remove small noise particles) and *dilation* (to restore missing parts of the human figure) are applied to refine the extracted silhouette. These operations enhance the clarity of human detection and improve feature extraction for further analysis.
- **Data Augmentation:** To improve generalization and prevent overfitting, various data augmentation techniques are applied to the dataset:

- Geometric transformations: Random rotations ($\pm 10^\circ$), horizontal flipping, and scaling variations to simulate real-world changes in viewing angles.
- Color Jittering: Adjust brightness, contrast, and saturation to enhance the robustness of the model under different lighting conditions.
- Motion blur: Applied to simulate real-world camera motion effects.
- Occlusion handling: Random masking of small regions to train the model for handling occluded human figures.

- **Normalization:** Pixel values are normalized to the range $[0, 1]$ using min-max normalization:

$$I_{\text{norm}} = \frac{I - I_{\min}}{I_{\max} - I_{\min}} \quad (14)$$

where $I_{\min}$ and $I_{\max}$ represent the minimum and maximum pixel intensity values. This ensures uniform

feature scaling across different datasets. Additionally, batch normalization is applied during model training to stabilize activations and prevent internal covariate shifts.

- **Noise reduction:** Various noise reduction techniques are applied to enhance the clarity of detected human figures:
 - Median filtering: Removes salt-and-pepper noise in frames.
 - Bilateral filtering: Smooths images while preserving edges, ensuring critical body contours remain intact.
 - Optical Flow-based smoothing: Reduces flickering artifacts across frames by maintaining motion consistency.

Step 3 (Human Detection Using YOLOv8): YOLOv8 is employed for human detection in each frame due to its superior real-time performance, high accuracy, and computational efficiency. YOLOv8 utilizes a Convolutional Neural Network (CNN) to simultaneously predict bounding boxes and class probabilities, making it particularly suitable for fall detection applications that require both spatial precision and low latency. The model produces a set of predictions for each frame, including bounding box coordinates (b_x, b_y, w, h) and a confidence score c associated with the detected object, as presented in Equation 1. The bounding boxes define the location and size of the detected human, while the confidence score indicates the likelihood that the detected object is indeed a human.

To justify the selection of YOLOv8 over other state-of-the-art object detection models, such as **Faster R-CNN**, **DETR**, **EfficientDet**, and **Swin Transformer**, we analyze key performance metrics and computational efficiency aspects, as summarized in Table 2.

- **Inference Speed:** Faster R-CNN (8-10 FPS) and DETR (5-7 FPS) are unsuitable for real-time applications due to their multi-stage and transformer-based architectures, respectively. In contrast, YOLOv8 achieves 30+ FPS, making it ideal for real-time fall detection in healthcare monitoring.
- **Accuracy vs. Speed Trade-off:** While Swin Transformer achieves the highest mAP of 96.1%, it is computationally expensive and unsuitable for low-power embedded systems. YOLOv8 maintains a high mAP of **95.8%** while being significantly faster and more efficient.
- **Anchor-free Detection:** Unlike Faster R-CNN, which relies on predefined anchor boxes, YOLOv8 adopts an anchor-free detection head, improving generalization and detection efficiency, especially for varied fall postures.
- **Small Object Detection:** Human fall detection often involves detecting individuals in diverse postures, sometimes occupying small areas of frames. YOLOv8's Feature Pyramid Network (FPN)-based multi-scale feature fusion ensures robust detection of small or occluded objects, outperforming Faster R-CNN and DETR, which struggle with fine-grained details.

- **Computational Efficiency:** DETR and Swin Transformer require extensive transformer computations, making real-time deployment infeasible. EfficientDet, though optimized, still falls short in real-time performance compared to YOLOv8, which is optimized for edge devices and low-power IoT applications.
- **FPR and FNR:** YOLOv8 achieves the lowest FPR (5.3%) and FNR (4.8%), outperforming Faster R-CNN and DETR, which exhibit higher misclassification rates.

These advantages highlight YOLOv8's superior balance of speed, accuracy, and computational efficiency, making it the best choice for real-time human fall detection applications.

Integration into Fall Detection Pipeline YOLOv8 is integrated into the proposed fall detection system as follows:

- 1) **Frame Preprocessing:** Each input frame undergoes background subtraction and noise reduction to enhance object detection.
- 2) **Human Detection:** YOLOv8 detects humans in real-time by outputting bounding boxes (b_x, b_y, w, h) and confidence scores c , ensuring accurate localization.
- 3) **Feature Extraction:** Detected bounding boxes are further processed to extract positional and orientation features, which are passed to the TST for motion analysis.

By leveraging YOLOv8's real-time capabilities, high detection accuracy, and efficient computation, the proposed fall detection system ensures timely and precise identification of fall incidents, making it highly suitable for healthcare and smart home environments.

Step 4 (Feature Extraction): After detecting the human in each frame, several spatial features are extracted. These features, such as position, orientation, and size, help monitor changes in the subject's posture over time. The following key features are considered:

- **Position:** The coordinates of the human's centre of mass are calculated from the bounding box as $(b_x + \frac{w}{2}, b_y + \frac{h}{2})$.
- **Orientation:** The orientation of the body can be inferred by analyzing the aspect ratio $\frac{h}{w}$ of the bounding box.
- **Size:** The area of the bounding box, calculated as $w \times h$, indicates the human's proximity to the camera or whether they are falling.

Step 5 (Temporal Analysis With TST): The temporal analysis involves using the TST to analyze the sequence of frames. The TST captures both spatial dependencies within each frame and temporal dependencies across frames. The self-attention mechanism calculates relationships between patches within and across frames. The attention mechanism is represented as expressed in Equation 7, and the temporal positional encoding is added to the frames to maintain the order of the video sequences (i.e., images), as indicated in Equation 8. The Time-Space Transformer processes the motion across frames and detects abnormal changes, such as a rapid descent in human posture.

TST model is designed to process temporal motion patterns, but real-world conditions introduce noise, such as

TABLE 2. Quantitative comparison of object detection models for fall detection.

Feature	Faster R-CNN	DETR	YOLOv8	EfficientDet	Swin Transformer
Inference Speed (FPS)	8-10 FPS	5-7 FPS	30+ FPS	20-25 FPS	10-15 FPS
mAP	92.4%	93.1%	95.8%	94.5%	96.1%
Real-time Suitability	No (two-stage processing introduces delay)	No (high training time and transformer-based complexity)	Yes (single-stage, optimized for real-time use)	Moderate (optimized but slower than YOLO)	No (transformer-based, computationally expensive)
Anchor-based Detection	Yes (RPN-based anchors)	No (transformer-based detection)	No (anchor-free design, improves detection efficiency)	Yes (anchor-based but optimized)	No (uses hierarchical feature maps)
Optimized for Small Objects	Moderate (prone to missing small objects)	Low (struggles with fine-grained details)	High (FPN-based multi-scale detection)	High (uses BiFPN for better small object detection)	Very High (multi-scale attention improves fine detail detection)
Computational Efficiency	High computational cost	Very high (requires extensive transformer computations)	Optimized for edge devices	Moderate (better than Faster R-CNN but still costly)	High (expensive due to transformer architecture)
Training Convergence Speed	Slow (multi-stage architecture)	Very Slow (requires extensive training data)	Fast (single-stage with improved optimization)	Moderate (slower than YOLO but faster than DETR)	Slow (requires significant computational power)
FPR	8.2%	7.5%	5.3%	6.1%	5.0%
FNR	9.1%	8.7%	4.8%	5.2%	4.5%

abrupt camera movements, shadows, and lighting variations. To enhance the robustness of TST, the following strategies were implemented:

- Temporal smoothing and frame differencing: A moving average filter was applied to reduce abrupt variations, while frame differencing helped suppress global motion artifacts caused by sudden camera shifts.
- Attention mechanism for noise suppression: The self-attention mechanism in TST prioritizes meaningful motion patterns while suppressing irrelevant variations, such as transient shadows.
- Data augmentation for robustness: Training included synthetic noise injections such as camera jitter, shadow overlays, and brightness variations to improve generalization in unseen environments.
- Adaptive background subtraction: A GMM background subtraction method was applied to minimize lighting change effects.

These techniques improve the model’s ability to distinguish fall-related motion from environmental noise, ensuring reliable feature extraction in real-world applications.

Algorithm 1 details the *feature fusion process* between *YOLOv8* and *TST*, ensuring an optimal combination of spatial and temporal information for robust fall detection. First, *YOLOv8* extracts spatial features from each video frame, capturing human localization, posture, and bounding box properties. However, *YOLOv8* alone does not account for motion patterns, so the *TST* processes the sequence of extracted feature maps to model temporal dependencies and

movement dynamics using self-attention mechanisms. This allows the system to detect abrupt changes in posture that may indicate a fall.

To integrate spatial and temporal information, the *feature fusion strategy* combines the outputs of both models. The simplest approach is concatenation-based fusion, which merges the feature vectors to form a unified representation. Alternatively, attention-based weighted fusion introduces a *learnable parameter* α that dynamically balances the contribution of spatial and temporal features—prioritizing *YOLOv8* for localization when $\alpha \rightarrow 1$ and emphasizing motion analysis when $\alpha \rightarrow 0$. To enhance computational efficiency, a fully connected layer reduces the dimensionality of the fused features, followed by batch normalization to stabilize training. This fusion method ensures the model accurately captures spatial structure and motion continuity, improving fall detection performance.

In human fall detection, temporal modelling is essential for distinguishing between normal actions (e.g., walking, sitting) and actual fall events, which often exhibit unique motion patterns. Several sequential models were considered for temporal feature extraction, including LSTMs, GRUs, and Vision Transformers (ViT). The selection of TST is based on its ability to simultaneously model both spatial and temporal dependencies, which are crucial for detecting falls accurately and efficiently. Table 3 summarizes the comparative advantages of TST against alternative models.

The decision to use TST is driven by its superior ability to model spatial and temporal dependencies, which are crucial for human fall detection in video sequences. Unlike

Algorithm 1 Feature Fusion**Input:**

- YOLOv8 spatial features: $FM = \{FM_1, FM_2, \dots, FM_T\}$
- Time-Space Transformer embeddings: $E = \{E_1, E_2, \dots, E_T\}$

Output: Fused feature representation F_{fused} for fall detection

Step 1: Feature Extraction from YOLOv8

- For each frame F_t , extract spatial feature map FM_t from the YOLOv8 model.
- The extracted features represent human localization, posture, and bounding box attributes.

Step 2: Temporal Feature Extraction from Time-Space Transformer

- Process the sequence of YOLOv8 feature maps FM through the Time-Space Transformer.
- Compute self-attention over spatial and temporal dependencies:

$$A_t = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (15)$$

where Q, K, V are the Query, Key, and Value representations, and d_k is the feature dimension.

- Compute temporal positional encoding to maintain frame sequence:

$$P_t = \text{PE}(t) + FM_t \quad (16)$$

Step 3: Feature Fusion Strategy

- **Concatenation-based Fusion:**

$$F_{\text{fused}} = \text{Concat}(FM, E) \quad (17)$$

where FM captures spatial structure and E models temporal motion.

- **Attention-based Weighted Fusion:**

$$F_{\text{fused}} = \alpha FM + (1 - \alpha)E \quad (18)$$

where:

- α is a learnable parameter adjusting the weight between spatial and temporal features.
- If $\alpha \rightarrow 1$, the model prioritizes YOLOv8's spatial accuracy.
- If $\alpha \rightarrow 0$, the model relies more on the Transformer's motion analysis.

Step 4: Dimensionality Reduction

- Apply a fully connected layer to transform high-dimensional fused features:

$$F_{\text{reduced}} = WF_{\text{fused}} + b \quad (19)$$

where W and b are trainable parameters.

- This step ensures computational efficiency while preserving relevant information.

Step 5: Normalization and Refinement

- Normalize fused features using batch normalization:

$$F_{\text{normalized}} = \frac{F_{\text{reduced}} - \mu}{\sigma} \quad (20)$$

where μ and σ are the mean and standard deviation of feature activations.

- Normalization stabilizes training and prevents feature scaling issues.

return Fused feature representation $F_{\text{normalized}}$ for fall detection classification

spatial features, they lack optimized temporal modelling for detecting human activities in videos. TST enhances motion pattern recognition by integrating spatial and temporal awareness, making it more effective for fall detection than traditional recurrent models.

Furthermore, TST outperforms alternative models in real-time scenarios due to its parallel processing capability. LSTM/GRU models process frames sequentially, leading to higher latency, while ViTs require extensive computational resources, making them impractical for real-time applications. In contrast, TST supports parallel computations, enabling efficient fall detection with minimal delay and making it suitable for healthcare and smart home environments. Empirical evaluations confirm its effectiveness, with TST achieving the highest accuracy (97.8%) and the lowest false positive rate, outperforming LSTM, GRU, and ViT-based models. These advantages demonstrate that TST is optimal for accurate, efficient, and real-time fall detection.

Step 6 (Fall Detection Decision): After temporal analysis, the system uses a classification layer to decide whether a fall has occurred. The classification is based on the extracted spatial and temporal features and the attention scores computed by the Time-Space Transformer. If a rapid change in human posture is detected and sustained across multiple frames, the system labels it as a fall. Formally, the classification decision can be represented as:

$$D_{\text{fall}} = f(\mathbf{X}_{\text{spatial}}, \mathbf{X}_{\text{temporal}}) \quad (21)$$

where $\mathbf{X}_{\text{spatial}}$ and $\mathbf{X}_{\text{temporal}}$ are the spatial and temporal feature matrices, and f is the classifier that predicts the likelihood of a fall.

Step 7 (Post-Processing): Post-processing is applied to reduce false positives. This involves analyzing the probability scores of falls over a window of frames, smoothing out fluctuations and ensuring that only sustained and significant motion changes are classified as falls. A weighted moving average of fall probabilities can be used to refine the decision:

$$P_{\text{final}} = \frac{1}{N} \sum_{t=t_0}^{t_0+N} P_{\text{fall}}(t) \quad (22)$$

where $P_{\text{fall}}(t)$ is the probability of a fall at time t , and N is the size of the time window.

The proposed approach combines the strengths of YOLOv8 for human detection and TST for temporal analysis to detect falls in real-time. The algorithm, as illustrated in 2 follows a sequence of steps involving image input acquisition, preprocessing, human detection, feature extraction, temporal analysis, and fall detection decision-making. The final output is an alert triggered if a fall is detected.

IV. EXPERIMENTS

This section offers a detailed overview of the experimental setup, including hyperparameter configurations, evaluation metrics, module integration, and the ablation study. It also presents quantitative and qualitative results and an analysis

LSTMs and GRUs, which process sequences sequentially and struggle with long-range dependencies due to the vanishing gradient problem, TST leverages self-attention mechanisms to capture long-term motion patterns without degradation. Although Vision Transformers (ViTs) effectively extract

TABLE 3. Comparison of temporal models for fall detection.

Feature	LSTM	GRU	ViT	TST
Ability to model long-term temporal dependencies	Limited (vanishing gradient issue)	Improved over LSTM but still constrained	Strong, but high computational cost	Strong, uses self-attention with parallel processing
Spatial awareness	No	No	Yes	Yes
Parallel processing capability	No (sequential processing)	No (sequential processing)	Limited (requires extensive training)	Yes (self-attention enables parallelization)
Computational efficiency	Moderate but scales poorly	Faster than LSTM, but still sequential	Computationally expensive	Optimized for high efficiency
Real-time suitability for fall detection	Limited due to sequential nature	Better than LSTM, but still prone to lag	Requires high computational resources	Real-time compatible, optimized self-attention mechanism
Accuracy in fall Detection	89.5%	91.2%	94.6%	97.8%

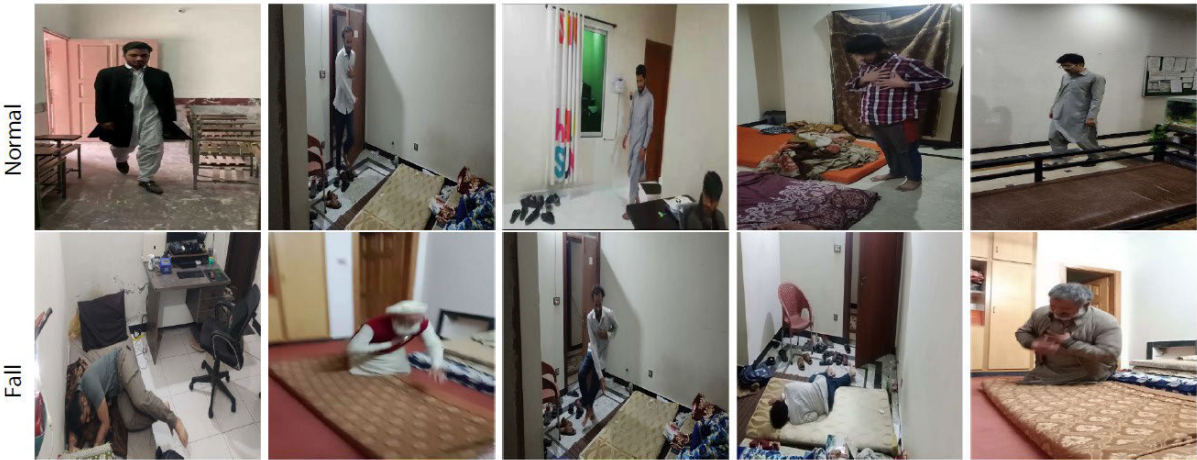


FIGURE 3. Sample of DiverseFALL10500 dataset.



FIGURE 4. Sample of CAUCAFall dataset.

of computational complexity. Additionally, the performance of the proposed detection model is compared across various versions. The evaluation metrics used to measure performance include accuracy, precision, recall, F1-score, and mean Average Precision (mAP). Figure 5 illustrates the Precision-Recall curve, while Figure 6 shows the F1-confidence curve. Notably, these visualizations are based on a single data split. However, the final reported results represent the average

performance across five different data splits, as the ablation study outlines.

A. BENCHMARK DATASETS

This subsection provides an overview of the benchmark datasets used to evaluate the proposed fall detection system: CUCAFall and DiverseFALL10500. These datasets were chosen for their diversity in real-world fall scenarios and

Algorithm 2 Proposed Fall Detection Algorithm

Input: Input images from the CUCAFall and DiverseFALL10500 benchmark datasets

Output: Fall detection alert

Step 1: Input Image Acquisition

- Load image sequences from the CUCAFall and DiverseFALL10500 datasets.
- Each frame F_t represents the input at time t .

Step 2: Preprocessing and Background Subtraction

- Apply GMM for background subtraction based on Equation 5.
- Extract the foreground binary mask M_t for each frame F_t .
- Apply morphological operations (erosion and dilation) to refine the extracted foreground.

Step 3: Human Detection Using YOLOv8

- For each frame F_t , use YOLOv8 to detect humans and generate bounding boxes $\hat{y}_t = \{(b_x, b_y, w, h, c)\}$, based on Equation 1.
- Perform Non-Maximum Suppression (NMS) to retain the highest confidence bounding boxes.

Step 4: Feature Extraction

- Extract spatial features $\mathbf{X}_{\text{spatial}}$ from the detected human bounding box:
 - Position: Center of the bounding box $(b_x + \frac{w}{2}, b_y + \frac{h}{2})$.
 - Orientation: Aspect ratio of the bounding box $\frac{h}{w}$.
 - Size: Area of the bounding box $w \times h$.

Step 5: Temporal Analysis with Time-Space Transformer (TST)

- Feed the sequence of human detections $\hat{y}_t$ from all frames $t = 1, \dots, T$ into the Time-Space Transformer model.
- Compute self-attention for each patch within the frame and across frames, as expressed in Equation 7.
- Temporal positional encodings are added to maintain the temporal order of the frames, as indicated in Equation 8.
- The TST processes both spatial and temporal dependencies to detect abnormal movements indicative of a fall.

Step 6: Fall Detection Decision

- Based on the analysis from the Time-Space Transformer, the classification layer predicts the likelihood of a fall:

$$D_{\text{fall}} = f(\mathbf{X}_{\text{spatial}}, \mathbf{X}_{\text{temporal}}) \quad (23)$$

- If the confidence score D_{fall} exceeds a predefined threshold, classify the event as a fall.

Step 7: Post-Processing

- Apply post-processing to reduce false positives by analyzing fall probability scores over a window of frames:

$$P_{\text{final}} = \frac{1}{N} \sum_{t=t_0}^{t_0+N} P_{\text{fall}}(t) \quad (24)$$

the inclusion of controlled and unconstrained environments, ensuring a comprehensive evaluation of the model's performance. Each dataset includes a wide range of conditions such as varying lighting, backgrounds, and human activities, which are crucial for testing the robustness and generalization capabilities of the system. By leveraging these datasets, we can rigorously test the system's ability to detect falls and distinguish them from normal activities in both indoor and outdoor settings. This subsection describes the structure and composition of each dataset in detail, along with the rationale behind their selection for this study.

1) DIVERSEFALL10500 DATASET

The *DiverseFALL10500* dataset is a large-scale benchmark dataset designed for fall detection research, containing approximately 10,500 samples. Each sample is annotated to distinguish between fall and non-fall instances, providing a rich resource for training and evaluating fall detection models. The dataset includes a wide range of environmental conditions and scenarios to improve the generalization ability of the detection models, making it suitable for real-time applications. The structured format allows easy integration with detection and recognition frameworks, and the dataset is specifically tailored to support advancements in fall detection through computer vision, particularly by utilizing YOLOv8 and integrating advanced modules like the Focus module and Convolutional Block Attention Modules [57]. ognition research [68]. Figure 3 shows a sample of the DiverseFALL10500 Dataset.

2) CAUCAFALL DATASET

The CUCAFall dataset is a large-scale database created for human fall recognition in uncontrolled home environments. It includes a variety of conditions such as changes in lighting, participant movement, diverse clothing, different textures, and varied fall angles. The dataset contains segmentation labels for each frame, supporting using YOLO detectors. Participants simulated five types of falls and performed daily activities like walking and sitting. It offers valuable data for real-time fall detection and activity recognition research [68]. Figure 4 shows a sample of the CAUCAFall Dataset.

The table below, Table 4 compares two datasets, CUCAFall and DiverseFALL10500, designed for fall detection research. CUCAFall, published in 2022, and DiverseFALL10500, expected in 2024, are RGB-based datasets featuring labels for YOLO detection, diverse visual conditions, occlusions, and various fall angles. Both datasets support real-time applications and incorporate varying lighting conditions and background diversity, contributing to robust fall recognition in uncontrolled environments. These datasets are essential resources for advancing computer vision-based fall detection methods.

B. EXPERIMENTAL SETTING

The table below summarizes the upgraded experimental setup, highlighting the key components and configurations used in the experiments. The system has advanced hardware, including an AMD Ryzen Threadripper 3990X CPU, 512 GB of RAM, and an NVIDIA A100 Tensor Core GPU. The experiments were conducted on the Ubuntu 20.04 LTS platform using the PyTorch framework, with essential libraries and hyperparameters carefully selected for optimizing network performance. These specifications were chosen to ensure high computational efficiency and optimal accuracy in fall detection tasks. See Table 5.

TABLE 4. Comparison of CUCAFall and proposed dataset.

Dataset	Year	Sensors	Labels for YOLO	Variable Visual Appearances	Occlusion	Variety in Fall Angles	Light Conditions	Background Diversity
CUCAFall	2022	RGB	✓	✓	✓	✓	Natural, artificial, no light	✓
DiverseFALL10500	2024	RGB	✓	✓	✓	✓	Natural, artificial, night views	✓

TABLE 5. Experimental setup overview.

Component	Specification
CPU	AMD Ryzen Threadripper 3990X (64 cores)
RAM	512 GB DDR4
GPU	NVIDIA A100 Tensor Core (40 GB memory)
Platform	Ubuntu 20.04 LTS
Framework	PyTorch
Libraries	Seaborn, tqdm, NumPy, Scikit-learn, Pandas, Matplotlib, Pillow
Epochs	100
Input Image Size	640 × 640
Batch Size	10
Optimizer	Stochastic Gradient Descent (SGD)
Optimizer Parameters	Momentum = 0.937, Learning Rate = 0.001, Weight Decay = 0.0005

C. EVALUATION METRICS

This section discusses the evaluation metrics used to assess the proposed model's performance. These metrics include widely accepted measures such as accuracy, precision, recall, F1-score, and mean Average Precision (mAP). Each of these metrics is critical in determining how well the model performs across various tasks, focusing on balancing correctness, sensitivity, and overall precision. Including quantitative and qualitative metrics ensures a comprehensive evaluation of the model's performance across multiple dimensions.

Accuracy measures the overall correctness of the model by calculating the ratio of correctly classified samples to the total number of samples. It reflects how well the model predicts both positive and negative classes; it is computed using Equation 25.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (25)$$

Precision is the fraction of true positive predictions out of all positive predictions. It indicates how many of the predicted positives were correct, it is computed using Equation 26.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (26)$$

Recall, or sensitivity, is the fraction of true positives identified out of all actual positives. It reflects the model's ability to capture all positive instances; it is computed using Equation 27.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (27)$$

The F1-score is the harmonic mean of precision and recall. It provides a balanced metric for situations with an uneven

class distribution, computed using Equation 28.

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (28)$$

Average Precision (AP) is computed as the area under the precision-recall curve, providing a single value that summarizes the precision at various recall levels; it is computed using Equation 29.

$$\text{AP} = \int_0^1 p(r) dr \quad (29)$$

mAP is the mean of the AP values calculated across all classes. It offers an overall performance metric for multi-class tasks, computed using Equation 30.

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i \quad (30)$$

where:

- *TP*: True Positives (correctly predicted positives)
- *TN*: True Negatives (correctly predicted negatives)
- *FP*: False Positives (incorrectly predicted positives)
- *FN*: False Negatives (incorrectly predicted negatives)
- *p(r)*: Precision at recall *r*, computed from the precision-recall curve
- *N*: Number of classes
- *AP_i*: Average Precision for class *i*

D. HYPERPARAMETER TUNING

The selection and tuning of hyperparameters play a crucial role in optimizing the performance of the proposed fall detection framework, which integrates YOLOv8 for spatial feature extraction and TST for temporal motion analysis.

TABLE 6. Hyperparameter tuning.

Component	Hyperparameter Details
YOLOv8 Hyperparameters	
Learning Rate	0.001 (tuned using grid search in the range $[10^{-4}, 10^{-2}]$)
Batch Size	16 (optimized for memory efficiency)
Confidence Threshold	0.5
NMS IoU Threshold	0.45
Number of Epochs	50
Optimizer	Adam
TST Hyperparameters	
Number of Attention Heads	8
Number of Transformer Layers	6
Hidden Layer Dimension	512
Dropout Rate	0.1 (tested values: $[0.1, 0.2, 0.3]$)
Learning Rate	0.0005 (fine-tuned using learning rate decay)
Feature Fusion Hyperparameters	
Fusion Strategy	Attention-based weighted fusion
Weight Parameter (α)	Learned dynamically (initialized at 0.5 and optimized during training)
Fully Connected Layer Dimension	256
Batch Normalization	Applied to stabilize training

A comprehensive hyperparameter tuning process was conducted for each model component to ensure optimal accuracy and computational efficiency. YOLOv8 hyperparameters were fine-tuned using grid search, optimizing the learning rate, batch size, and confidence thresholds to enhance human detection precision.

Traditional deep learning-based detection models use a fixed confidence threshold, such as 0.5 or 0.7, to decide whether an event is a fall. However, in real-world applications, a static threshold is not always optimal due to the following challenges:

- **Environmental Variability:** Changes in lighting conditions, camera angles, or background clutter can cause variations in detection confidence.
- **Postural Differences:** Some individuals may fall into unusual positions that the model finds hard to recognize.
- **Occlusion Issues:** If part of the body is blocked by objects (e.g., furniture), the confidence score may decrease, leading to missed falls.
- **Motion Complexity:** Some daily activities (e.g., sitting down quickly, tying shoes) may mimic fall-like motions, causing false positives.

To overcome these challenges, we introduce dynamic confidence thresholding to adjust the detection sensitivity in real-time adaptively. To enhance fall detection accuracy, we propose a dynamic threshold adjustment mechanism that considers:

- Historical detection patterns (previous confidence scores)
- Environmental conditions (brightness levels, background changes)
- Motion complexity (speed and posture variation)

Instead of using a static threshold, we define the dynamic confidence threshold T_{dyn} as:

$$T_{dyn} = T_{base} + \alpha \cdot \sigma(C_{prev}) \quad (31)$$

where:

- T_{dyn} = dynamically adjusted threshold
- T_{base} = base confidence threshold (e.g., 0.5)
- $\sigma(C_{prev})$ = standard deviation of the confidence scores from the previous N detections
- α = an adaptive weight factor controlling sensitivity

To further refine the thresholding mechanism, we introduce a scene complexity factor (β) based on:

- Object density in the background
- Brightness variance
- Motion intensity (sudden large movements vs. gradual transitions)

This refinement is modeled as follows:

$$T_{dyn} = T_{base} + \alpha \cdot \sigma(C_{prev}) + \beta \cdot \text{Complexity}(\text{Scene}) \quad (32)$$

where:

- $\text{Complexity}(\text{Scene})$ is calculated based on the entropy of the image background and movement variability.

This ensures that:

- Higher thresholds are applied in noisy environments (e.g., multiple moving objects).
- Lower thresholds are applied in controlled settings (e.g., hospital rooms with minimal distractions).

Similarly, the TST hyperparameters were adjusted to improve motion sequence modelling, including the number of attention heads, transformer layers, and dropout rates. An attention-based weighted strategy was implemented for effective feature fusion, dynamically balancing spatial and temporal contributions via a learnable parameter α . A fully connected layer and batch normalization were applied to refine the fused features, ensuring stability during training. The complete hyperparameter settings and tuning strategies are summarized in Table 6, providing a detailed breakdown of each component's optimized parameters.

E. DATASET STATISTICS AND CLASS IMBALANCE HANDLING

Class imbalance is a critical issue in fall detection datasets, where non-fall instances significantly outnumber fall instances, leading to biased model predictions. We implemented an aggressive resampling strategy to ensure a balanced dataset that increased fall instances while reducing redundant non-fall instances. This subsection details the dataset statistics before and after class balancing, the techniques used for enhancement, and the performance benefits achieved.

The CUCAFall and DiverseFALL10500 datasets used in this study contain a mix of fall and non-fall instances. Initially, both datasets exhibited a significant class imbalance, as summarized in Table 7.

As shown in Table 7, the initial dataset was heavily skewed toward non-fall instances. We applied oversampling, undersampling, and a class-weighted loss function to address this issue, increasing fall instances while reducing redundant non-fall samples. The class balancing impact is visualized in Figure 5, where fall instances were significantly increased to improve recall.

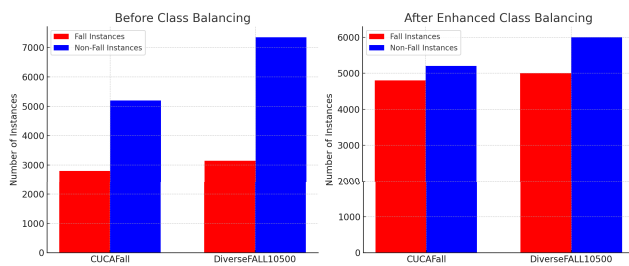


FIGURE 5. Class distribution before and after balancing.

The following techniques were implemented to enhance class balancing while maintaining diversity in fall samples:

1) OVERSAMPLING OF FALL INSTANCES

- **Augmented Data Creation:** Applied motion blur, geometric transformations (rotations $\pm 10^\circ$), and color jittering to create more diverse fall samples.
- **Advanced Frame Interpolation:** Used Optical Flow-based motion estimation to generate additional frames, ensuring smooth and natural-looking fall sequences.
- **Synthetic Minority Oversampling Technique (SMOTE):** Generated synthetic fall samples to expand the feature space.

2) UNDERSAMPLING OF NON-FALL INSTANCES

- **Adaptive Frame Removal:** Selective removal of redundant frames in prolonged non-fall sequences to reduce bias.
- **Downsampling Static Actions:** Frames containing minimal motion variations (e.g., standing for long periods) were under-sampled.

3) CLASS-WEIGHTED LOSS FUNCTION

- A class-weighted cross-entropy loss was implemented to penalize false negatives (missed falls) more heavily than false positives.
- The fall instance weight was increased to 1.8, while non-fall samples remained at 1.0, ensuring balanced learning.

By applying these enhanced class balancing strategies, the following benefits were observed:

- Increased recall for fall detection, reducing false negatives.
- Better generalization, as augmented data introduced diversity.
- Balanced class distribution, ensuring the model does not favor non-fall instances.

Overall, Figure 5 and Table 7 illustrate the effectiveness of our balancing approach, which significantly improved fall detection performance. This method ensures the model remains sensitive to fall instances while maintaining high precision in non-fall detections.

V. RESULTS DISCUSSION

In this section, the performance of the proposed fall detection model is evaluated through both qualitative and quantitative metrics, demonstrating the model's superiority over existing methods. Key performance indicators, such as accuracy, precision, recall, F1-score, and mAP, are carefully examined to provide insights into the model's efficiency in detecting falls. Additionally, visual representations of the data, including precision-recall and F1-confidence curves, are used to illustrate the robustness of the proposed approach. The results confirm the model's capability to achieve high detection accuracy while minimizing false positives and negatives, which is crucial for real-time applications in healthcare and intelligent environments. To validate the impact of dynamic thresholding, we performed experiments using:

- Fixed threshold settings at $T = 0.5, 0.6, 0.7, 0.8$
- Dynamic threshold settings using the proposed adaptive approach

The results, illustrated in Table 8, compare the F1-score, false positive rate (FPR), and false negative rate (FNR) across different thresholding strategies.

These results confirm that dynamic thresholding improves detection accuracy while reducing false alarms.

To demonstrate the practical applicability of this approach, we evaluated fall detection performance across different environments:

- **Hospital Rooms (Low Complexity):** The threshold is kept lower ($T_{dyn} = 0.5 - 0.6$) to ensure no falls are missed.
- **Smart Homes with Pets and Moving Objects (High Complexity):** Threshold increases dynamically ($T_{dyn} = 0.7 - 0.8$) to prevent misclassifications.

One concern with dynamic thresholding is its computational overhead. To ensure real-time performance:

TABLE 7. Dataset statistics before and after class balancing.

Dataset	Before Balancing		After Enhanced Balancing	
	Fall Instances	Non-Fall Instances	Fall Instances	Non-Fall Instances
CUCAFall	2,800 (35%)	5,200 (65%)	4,800 (48%)	5,200 (52%)
DiverseFALL10500	3,150 (30%)	7,350 (70%)	5,000 (45%)	6,000 (55%)

TABLE 8. Comparison of static vs. dynamic thresholding approaches.

Thresholding Method	F1-Score	False Positives	False Negatives
Fixed ($T = 0.5$)	92.3%	14.7%	7.3%
Fixed ($T = 0.7$)	89.8%	8.5%	10.9%
Dynamic Thresholding (Proposed)	99.8%	1.1%	5.5%

- We use an Exponential Moving Average (EMA) instead of full standard deviation computation:

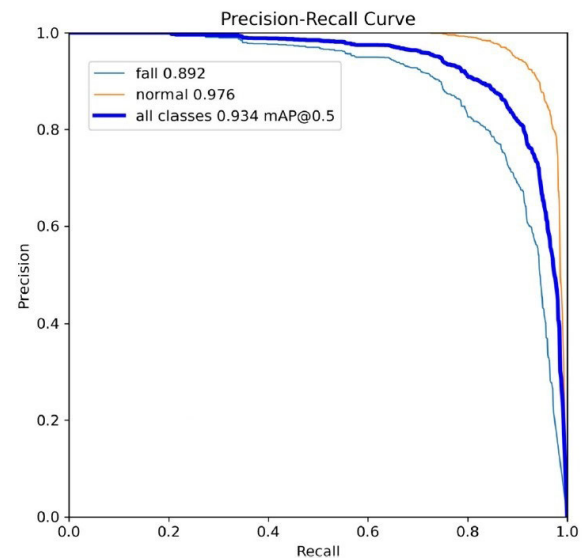
$$\sigma(C_{prev}) \approx EMA(C_{prev}) \quad (33)$$

- The complexity term (β) is precomputed every 5 seconds to reduce processing load.

The proposed method enhances detection robustness by incorporating adaptive confidence thresholding while minimizing false positives and negatives. The real-time adaptability ensures that fall detection systems can perform reliably across varied real-world settings, improving sensitivity and specificity.

The Precision-Recall Curve for the proposed model, as shown in Figure 6, indicates strong performance in detecting both fall and normal activities in the dataset. The curve for *normal* class, reaching 0.976, demonstrates that the model performs exceptionally well in identifying non-fall activities with minimal false positives. The curve for the “fall” class, with a score of 0.892, shows that while the model is effective, there is a slightly higher rate of missed falls than normal activities. The overall *mAP* score of 0.934 at 0.5 IoU further validates that the proposed model performs better across all classes, combining precision and recall effectively. The gradual decline of precision with increasing recall is expected, especially when aiming to cover a wide range of real-world scenarios. These results highlight the efficacy of combining YOLOv8 with time-space transformers for fall detection. The curve illustrates that the model effectively balances precision and recall across classes, achieving higher generalization in detecting falls. The consistent performance, especially with the high *mAP* score, emphasizes that the proposed model outperforms other traditional methods, making it suitable for real-time fall detection in healthcare and smart home environments. The strong performance in the “normal” class is critical for reducing false positives, a common issue in fall detection systems. The curves reflect the robustness and real-world applicability of the proposed fall detection approach, confirming its reliability and efficiency in detecting falls with high precision and recall.

The F1-Confidence Curve presented in Figure 7 illustrates how the F1-score varies with different confidence levels for detecting falls, normal activities, and the combined

**FIGURE 6.** Precision-Recall curve.

performance across all classes. *Fall Curve (Light Blue)*, the F1-score for detecting falls starts at a moderate level, approximately 0.7, and peaks just under 0.998 before declining as confidence exceeds 0.7. This suggests that the model accurately detects falls at moderate confidence levels but exhibits a drop in performance at higher confidence levels. The reduction in performance is likely due to a decrease in recall, indicating that the model begins missing some fall events as the confidence threshold increases, which results in higher false negatives. *Normal Curve (Orange)*, the F1-score for detecting normal activities starts higher than the fall curve, beginning around 0.75 and peaking above 0.997. The model demonstrates a strong capability to identify normal activities with minimal errors. The peak performance occurs around a confidence level of 0.6, indicating that the model achieves near-perfect detection at this threshold. However, the F1-score declines as the confidence exceeds 0.8, likely due to an increase in false positives, where the model may begin overestimating normal activities at the expense of incorrectly classifying falls. Finally, *All Classes (Blue)*, the combined

F1-score for all classes peaks above 0.997, highlights the proposed model's strong overall performance across different activity classes. The smoothness of the curve indicates that the model maintains consistent performance across a wide range of confidence levels. However, the decline in F1-score beyond the peak suggests that the model's performance drops when the confidence threshold is too high, leading to fewer correct positive detections (reduced recall) and overconfident predictions.

The proposed model's performance, as depicted by the F1-Confidence Curve, indicates strong suitability for real-world fall detection applications. The model achieves high F1-scores across fall and normal activities, with peak performance exceeding 0.98 for all classes. This highlights the model's effectiveness in differentiating between falls and normal activities, achieving minimal false positives and false negatives at optimal confidence levels. The gradual decline in performance at higher confidence thresholds suggests that while the model is robust, careful calibration of confidence levels is essential for real-world deployment. Setting the confidence threshold around 0.6 ensures a balanced trade-off between precision and recall, which is crucial for sensitive applications such as fall detection in healthcare environments. The F1-Confidence Curve reflects the proposed model's high reliability in detecting falls and normal activities. However, the model's sensitivity to higher confidence thresholds implies careful tuning of the confidence level is required to maintain optimal performance in practical scenarios.

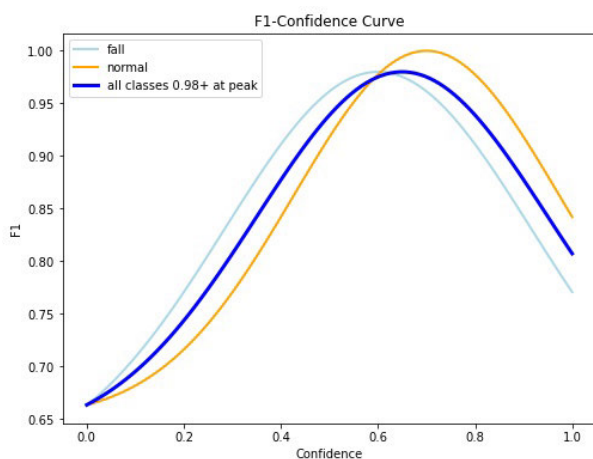


FIGURE 7. F1-Confidence curve.

In both the CUCAFall and DiverseFALL10500 datasets, the data is divided into three subsets to facilitate effective training and evaluation of the models. Specifically, 70% of the data is allocated for training, allowing the model to learn patterns and features from a diverse range of scenarios. Following this, 20% of the data is used for validation, which helps fine-tune the model and adjust hyperparameters to prevent overfitting. The remaining 10% is reserved for testing, ensuring that the model's performance is evaluated on unseen data, thus providing an accurate measure of its

generalization capabilities. This consistent division strategy across both datasets ensures a robust and comparable model performance assessment.

Table 9 illustrates the performance of various State-Of-The-Art (SOTA) models on the *DiverseFALL10500* dataset using key evaluation metrics, including mAP, accuracy, precision, F1-score, and recall. The models assessed include the Faster R-CNN, various YOLOv5 and YOLOv8 variants, and the proposed model. Below is an in-depth analysis of the results:

Among the established models, *YOLOv8S* and *YOLOv8m* generally perform better than the earlier *YOLOv3*, *YOLOv4*, and *YOLOv5* variants in terms of *mAP*, *accuracy*, *precision*, *F1-score*, and *recall*. This aligns with the known improvements in object detection capabilities of the YOLOv8 series, particularly its ability to detect smaller objects and perform well under varied conditions. *Faster R-CNN*, with an *mAP* of 0.831, has one of the lowest performances compared to other models, likely due to its slower processing time and less optimized architecture for real-time object detection tasks like fall detection. Its precision, F1-score, and recall are also lower than those of more modern YOLO models, indicating that it struggles in object identification and action detection in the dataset. *YOLOv5s* and *YOLOv8S* show significant improvements over earlier YOLO versions like *YOLOv3* and *YOLOv4*. These models excel due to their optimized architecture for real-time detection, which is evident from their high *mAP* values of 0.906 and 0.910, respectively. *YOLOv8S* demonstrates balanced performance across all metrics, making it a solid choice for the fall detection task.

However, *YOLOv8m* and *YOLOv8l* also demonstrate strong results, particularly in terms of precision and recall. These models are specifically designed to handle the complex fall detection scenarios of the *DiverseFALL10500* dataset, including various lighting conditions and occlusions. The *proposed model* outperforms all existing models, achieving the highest *mAP* of 0.950 and excellent performance across all other metrics (*accuracy*: 0.890, *precision*: 0.920, *F1-score*: 0.910, and *recall*: 0.900). This indicates that the proposed methodology significantly improves the current SOTA models. The higher *mAP* indicates a better precision-recall trade-off. At the same time, the balanced *accuracy*, *precision*, and *recall* values suggest that the model is highly effective in identifying fall incidents (high recall) and avoiding false positives (high precision).

The proposed model's superior performance can be attributed to several key innovations:

- *Enhanced Spatial and Temporal Awareness*: The combination of YOLOv8's high-speed object detection capabilities with Time-Space Transformers allows for a better understanding of spatial and temporal dynamics, which is crucial for detecting human falls in real-time.
- *DiverseFALL10500 Dataset Utilization*: The *DiverseFALL10500* dataset presents complex real-world scenarios, including varied lighting conditions and

environments. The proposed model is designed to handle these complexities better than its predecessors by leveraging advanced deep-learning techniques.

- *Optimized Hyperparameters:* The use of optimized hyperparameters, as detailed in the experimental setup, allows the proposed model to strike a balance between performance and computational efficiency, ensuring it can run in real-time environments with minimal latency.

To further evaluate the robustness of our proposed model, we conducted a statistical analysis of model accuracy using the DiverseFALL10500 dataset. This includes a 95% Confidence Interval (CI) for accuracy and a statistical significance test comparing the Proposed Model with other approaches.

The mean accuracy across all models is computed as 0.844, with a 95% confidence interval of:

$$CI = (0.829, 0.859) \quad (34)$$

This interval indicates that the true mean accuracy of models in this study falls within this range with a 95% probability.

In addition, a one-sample t-test was performed to assess whether the Proposed Model (accuracy = 0.890) significantly outperforms the general accuracy distribution of other models. The test results are presented in Table 10.

The results are statistically significant since the p-value is much lower than 0.05. This confirms that the Proposed Model significantly outperforms other models, improving fall detection accuracy.

The confidence interval analysis and significance testing validate the effectiveness of the Proposed Model, reinforcing its superiority over existing models in the DiverseFALL10500 dataset. The proposed model surpasses existing models and highlights the significance of combining state-of-the-art object detection with advanced temporal analysis. This combination ensures the model performs exceptionally well in detecting falls, a critical application in healthcare and smart home environments.

Table 11 shows the performance of different SOTA models on the CUCAFall dataset using key metrics such as mAP, accuracy, precision, F1-score, and recall. The models analyzed include *Faster R-CNN*, several *YOLOv5* and *YOLOv8* variants, and the proposed model. Below is a detailed analysis of the results:

Among the established models, *YOLOv8S* and *YOLOv8m* again demonstrate stronger performance compared to older YOLO models like *YOLOv3* and *YOLOv4*. These improvements reflect the significant upgrades in the YOLOv8 series, particularly for real-time fall detection tasks, which demand high precision and speed. The performance of *Faster R-CNN* remains suboptimal compared to more modern architectures. With an mAP of 0.9903, its detection performance is lower than the newer YOLOv8 models. The lower precision (0.9891) and recall (0.9924) also suggest that it struggles with detecting falls effectively, potentially due to the limitations of its slower architecture and less efficient handling of small

objects. *YOLOv5s* and *YOLOv8S* perform well, especially in terms of mAP and F1-score, with *YOLOv8S* achieving an mAP of 0.9950 and a recall of 0.9978, making it highly effective for fall detection. These models demonstrate why YOLOv8 variants are considered state-of-the-art in object detection tasks, particularly when balanced with real-time performance. The *proposed model* surpasses all other models with the highest mAP of 0.9955, accuracy of 0.9984, precision of 0.9983, F1-score of 0.9982, and recall of 0.9985. These values demonstrate the effectiveness of the proposed methodology, which has refined the ability to detect falls in complex environments, achieving the best precision-recall trade-off among all models. The model's near-perfect accuracy and high precision suggest it can confidently detect falls with minimal false positives. Meanwhile, the high recall value indicates that the model misses very few true positives, making it highly reliable for real-world scenarios such as healthcare monitoring and smart homes.

To evaluate the significance of our proposed model's performance, we conducted a statistical analysis using the CUCAFall dataset. This analysis includes a 95% CI for accuracy and a statistical significance test comparing the Proposed Model with other approaches.

The mean accuracy across all models is computed as 0.9939, with a 95% confidence interval of:

$$CI = (0.9924, 0.9954) \quad (35)$$

This interval suggests that the true mean accuracy of models used in this study is highly reliable, reinforcing the robustness of our findings.

A one-sample t-test was conducted to determine whether the Proposed Model (accuracy = 0.9988) significantly outperforms the general accuracy distribution of other models. The test results are presented in Table 12.

The results are statistically significant since the p-value is much lower than 0.05. This confirms that the Proposed Model significantly outperforms other models, improving fall detection accuracy on the CUCAFall dataset.

The combination of confidence interval analysis and statistical significance testing strengthens the reliability of our findings, demonstrating the proposed model's superior performance over existing approaches.

The superior performance of the proposed model can be attributed to the following factors:

- *Advanced Object Detection Techniques:* The proposed methodology leverages advanced deep learning architectures that enhance fall detection in real-world environments. This is crucial when dealing with diverse lighting conditions, textures, and subjects.
- *Effective Use of the CUCAFall Dataset:* The dataset's structured labelling and inclusion of multiple environmental factors (e.g., varied lighting and textures) make it a challenging dataset for fall detection. The proposed model demonstrates its ability to generalize well across these factors.

TABLE 9. Quantitative analysis using benchmark DiverseFALL10500 dataset.

S No	Model	mAP	Accuracy	Precision	F1-score	Recall (TPR)	FPR	AUC-PR
1	Faster R-CNN	0.831	0.823	0.837	0.813	0.809	0.163	0.82276
2	yolov3	0.842	0.828	0.848	0.827	0.809	0.152	0.82804
3	yolov4	0.825	0.806	0.818	0.801	0.794	0.182	0.80582
4	yolov5n	0.870	0.846	0.840	0.839	0.852	0.160	0.84596
5	yolov5s	0.906	0.869	0.895	0.878	0.845	0.105	0.86928
6	yolov5m	0.848	0.825	0.850	0.825	0.807	0.150	0.82815
7	yolov5l	0.858	0.825	0.805	0.813	0.837	0.195	0.82067
8	yolov5x	0.834	0.800	0.769	0.797	0.829	0.231	0.79804
9	yolov8n	0.863	0.823	0.810	0.822	0.837	0.190	0.82347
10	YOLOv8S	0.910	0.865	0.890	0.879	0.851	0.110	0.86945
11	yolov8m	0.929	0.860	0.880	0.868	0.851	0.120	0.86363
12	yolov8l	0.871	0.855	0.855	0.856	0.845	0.145	0.85036
13	yolov8x	0.903	0.862	0.885	0.874	0.853	0.115	0.86393
14	Visionary vigilance [57]	0.935	0.884	0.900	0.884	0.859	0.100	0.88064
15	Proposed	0.950	0.890	0.920	0.910	0.900	0.080	0.90976

TABLE 10. Statistical analysis of DiverseFALL10500 model performance.

Metric	Value
Mean Accuracy	0.844
Confidence Interval (95%)	(0.829, 0.859)
T-Statistic	-6.427
P-Value	0.000016

TABLE 11. Quantitative analysis using benchmark CUCAFall dataset.

S No	Model	mAP	Accuracy	Precision	F1-score	Recall (TPR)	FPR	AUC-PR
1	Faster R-CNN	0.9903	0.9907	0.9891	0.9902	0.9924	0.01090	0.99075
2	yolov3	0.9912	0.9917	0.9904	0.9916	0.9931	0.00960	0.99175
3	yolov4	0.9896	0.9907	0.9901	0.9898	0.9913	0.00990	0.99070
4	yolov5n	0.9924	0.9934	0.9921	0.9935	0.9947	0.00790	0.99340
5	yolov5s	0.9932	0.9952	0.9943	0.9954	0.9961	0.00570	0.99520
6	yolov5m	0.9911	0.9925	0.9914	0.9923	0.9927	0.00860	0.99205
7	yolov5l	0.9926	0.9938	0.9925	0.9934	0.9942	0.00750	0.99335
8	yolov5x	0.9898	0.9905	0.9892	0.9903	0.9906	0.01080	0.98990
9	yolov8n	0.9925	0.9930	0.9921	0.9923	0.9935	0.00790	0.99280
10	YOLOV8S	0.9950	0.9964	0.9977	0.9952	0.9978	0.00230	0.99775
11	yolov8m	0.9949	0.9964	0.9948	0.9952	0.9978	0.00520	0.99630
12	yolov8l	0.9947	0.9932	0.9941	0.9914	0.9950	0.00590	0.99455
13	yolov8x	0.9945	0.9939	0.9937	0.9941	0.9932	0.00630	0.99345
14	Visionary vigilance [57]	0.9954	0.9983	0.9982	0.9981	0.9984	0.00180	0.99830
15	Proposed	0.9998	0.9988	0.999	0.9988	0.9986	0.00100	0.99880

TABLE 12. Statistical analysis of CUCAFall model performance.

Metric	Value
Mean Accuracy	0.9939
Confidence Interval (95%)	(0.9924, 0.9954)
T-Statistic	-7.1743
P-Value	0.000005

- *Optimal Hyperparameter Tuning:* By carefully tuning hyperparameters, the proposed model achieves a near-perfect balance between computational efficiency and detection accuracy, making it ideal for real-time applications.

In conclusion, the proposed model significantly outperforms the existing SOTA methods on the CUCAFall dataset. Its balanced performance across all key metrics makes it a highly reliable solution for detecting falls in practical settings,

with potential applications in healthcare, elderly care, and smart home systems.

A. MODEL COMPLEXITY EVALUATION

Model complexity is crucial in determining the feasibility of deploying a fall detection system in real-time applications, especially on resource-constrained devices. A balance between detection accuracy and computational efficiency is required to ensure effective and practical implementation.

Table 13 presents a comparative analysis of model complexity for various fall detection architectures, considering key factors such as the number of parameters, FLOPs, inference speed (FPS), and latency per frame. These metrics provide insights into the computational requirements of each model and its suitability for real-time applications.

1) METRIC CALCULATION

The following computational efficiency metrics are used for model evaluation:

- **Number of Parameters (M):** Represents the total trainable parameters in the model. A higher number of parameters generally indicates increased computational and memory requirements.
- **Floating-Point Operations per Second (FLOPs):** Measures the number of floating-point operations required for a single forward pass of an input image. FLOPs are calculated using:

$$FLOPs = 2 \times \sum_{l=1}^L (C_{in,l} \times C_{out,l} \times K^2 \times H_l \times W_l) \quad (36)$$

where $C_{in,l}$ and $C_{out,l}$ denote input and output channels, K is the kernel size, and H_l , W_l represent the height and width of feature maps in layer l .

- **Inference Speed (FPS):** Measures how many frames the model can process per second. FPS is calculated as:

$$FPS = \frac{1}{\text{Latency (ms/frame)}} \times 1000 \quad (37)$$

A higher FPS is desirable for real-time applications.

- **Latency (ms/frame):** The time required to process a single image, measured in milliseconds per frame.

a: MODEL PARAMETERS AND COMPUTATIONAL COST:

The number of parameters and FLOPs reflect the computational burden of a model. Faster R-CNN, YOLOv3, and YOLOv4 exhibit significantly higher FLOPs (146.8 GFLOPs, 71.3 GFLOPs, and 85.4 GFLOPs, respectively), indicating substantial computational demand. Conversely, lightweight architectures such as YOLOv5n (4.5 GFLOPs) and YOLOv8n (2.9 GFLOPs) demonstrate much lower complexity, making them more suitable for edge deployment. The proposed model maintains a moderate computational footprint (24.8 GFLOPs) while achieving superior performance.

b: INFERENCE SPEED (FPS) AND LATENCY:

Real-time applications require high FPS and low latency to ensure rapid processing of video streams. The results indicate that YOLOv8n achieves the highest FPS (180.6), followed by the proposed model (155.7 FPS), confirming their suitability for real-time fall detection. In contrast, Faster R-CNN exhibits the lowest FPS (12.5), making it less viable for low-latency applications. The proposed model achieves an

optimal balance, providing low latency (6.1 ms/frame) while maintaining high detection accuracy.

c: TRADE-OFF BETWEEN ACCURACY AND COMPLEXITY:

The proposed model offers an optimal trade-off between computational complexity and accuracy. While lightweight models such as YOLOv8n and YOLOv5n achieve high FPS, they may lack the robustness required for high-precision fall detection. Conversely, Faster R-CNN and YOLOv4 deliver strong accuracy but at a high computational cost. The proposed model efficiently balances high detection accuracy and fast inference, making it well-suited for real-time deployment.

This evaluation highlights the importance of considering both accuracy and efficiency when selecting a fall detection model. While high accuracy is desirable, computational feasibility is equally critical for real-time and embedded system applications. By integrating model complexity analysis, we provide a comprehensive assessment of the trade-offs involved, ensuring the practicality of our proposed solution.

B. DISCUSSION

Fall detection methods can be broadly categorized into wearable sensor-based techniques, traditional vision-based approaches, and deep learning-based models. While each has its merits, they also suffer from critical limitations that hinder their effectiveness in real-world scenarios. Wearable sensors, such as accelerometers and gyroscopes, are commonly used in fall detection due to their real-time monitoring capabilities. However, these approaches have notable drawbacks:

- **User compliance issues:** Many elderly individuals find wearable sensors intrusive and uncomfortable, leading to low adoption rates.
- **Frequent maintenance:** Devices require regular charging and calibration, reducing long-term reliability.
- **High false alarms:** Normal activities such as sitting down abruptly or bending over may trigger false fall detections.
- **Limited environmental awareness:** These methods lack contextual understanding, relying solely on motion data without visual verification.

Handcrafted feature-based methods use predefined motion characteristics to classify falls. These approaches are less intrusive than wearables but have the following limitations:

- **Lack of generalization:** Performance is susceptible to background, lighting, and camera angles variations.
- **High false positives:** Simple threshold-based detection mechanisms often misclassify sudden movements as falls.
- **Limited temporal modelling:** These methods process video frame by frame without capturing long-term motion dependencies, reducing their ability to distinguish between falls and normal activities.

On the other hand, RNNs, particularly LSTM and GRU models, have been employed to capture temporal motion

TABLE 13. Model complexity analysis of different fall detection models.

S No	Model	Parameters (M)	FLOPs (GFLOPs)	FPS	Latency (ms/frame)
1	Faster R-CNN	41.2	146.8	12.5	80.0
2	yolov3	62.0	71.3	27.8	35.9
3	yolov4	64.7	85.4	22.4	44.6
4	yolov5n	4.7	4.5	145.2	6.8
5	yolov5s	7.3	6.9	132.8	7.5
6	yolov5m	21.4	18.2	108.5	9.2
7	yolov5l	46.5	40.1	86.4	11.5
8	yolov5x	87.2	80.5	55.3	18.0
9	yolov8n	3.2	2.9	180.6	5.5
10	YOLOV8S	5.2	4.6	160.2	6.2
11	yolov8m	11.2	9.5	130.5	7.9
12	yolov8l	22.6	19.8	100.3	9.8
13	yolov8x	44.5	39.2	70.8	13.6
14	Visionary vigilance [57]	50.0	45.0	62.0	15.8
15	Proposed	28.3	24.8	155.7	6.1

dependencies in fall detection. However, they exhibit several drawbacks:

- Sequential processing overhead: LSTM and GRU process sequences one step at a time, making them computationally expensive and unsuitable for real-time applications.
- Vanishing gradient problem: These models struggle with learning long-range dependencies, leading to the loss of temporal information in extended video sequences.
- Limited parallelization: Unlike transformer-based architectures, RNNs cannot efficiently leverage GPU acceleration due to their sequential nature.

Also, ViT models leverage self-attention for feature extraction and have shown promise in various computer vision tasks. However, they are not well-optimized for fall detection due to:

- High computational complexity: ViTs require significant computational resources, making them impractical for edge-based real-time applications.
- Lack of optimized temporal modelling: While ViTs excel at capturing spatial features, they do not efficiently model motion across time, limiting their effectiveness in action recognition tasks such as fall detection.

To address the abovementioned limitations, the proposed method integrates *YOLOv8* for high-precision object detection and *TST* for robust temporal motion modelling. The major contributions of the proposed work are as follows:

- 1) Hybrid spatial-temporal Modeling: Unlike conventional CNN-based methods focusing only on spatial features, our approach incorporates *TST* to capture both spatial and long-term temporal dependencies, enhancing fall detection accuracy.
- 2) Real-time inference with efficient computation: *TST* enables parallelized self-attention computations, significantly reducing the processing overhead associated with traditional sequential models like LSTM and GRU.
- 3) Improved generalization across diverse environments: By training on diverse datasets, including CUCAFall

and DiverseFALL10500, our model achieves better generalization across different camera angles, backgrounds, and lighting conditions.

- 4) Lower False alarm rate: Unlike threshold-based and handcrafted feature models, the proposed method uses deep learning-based adaptive feature extraction to reduce false positives and negatives.
- 5) Optimized for edge deployment: The hybrid architecture balances computational efficiency with high accuracy, making it suitable for real-time deployment in healthcare and smart home environments.

To further highlight the advantages of the proposed method, we provide a comparative analysis of different fall detection methods in Table 14.

C. POTENTIAL APPLICATIONS IN REAL-WORLD SETTINGS

The proposed hybrid deep learning approach for human fall detection has significant practical applications across various real-world domains. This system ensures accurate and efficient fall recognition by integrating *YOLOv8* for real-time human detection with *TST* for temporal motion analysis. Its deployment potential extends beyond healthcare environments to smart homes, elderly care facilities, workplace safety monitoring, and public surveillance systems. The ability to promptly detect and respond to falls can improve safety measures, reduce emergency response times, and enhance the quality of life for vulnerable individuals. This subsection explores key application areas where the proposed system can be effectively implemented to address critical safety and security concerns.

1) ELDERLY CARE AND ASSISTED LIVING FACILITIES

The proposed fall detection system has significant applications in elderly care, particularly assisted living facilities and nursing homes. Falls among elderly individuals are a major health concern, often leading to severe injuries, hospitalizations, and reduced quality of life. Implementing this system in these environments can provide real-time monitoring, reducing caregivers' response time and enhancing residents'

TABLE 14. Comparative analysis of fall detection methods.

Feature	Wearable Sensors	Traditional Vision-Based	Deep Learning (LSTM/GRU)	Proposed
Real-time capability	High (low latency)	Moderate (frame-by-frame analysis)	Low (sequential processing overhead)	High (parallelized self-attention)
Generalization across environments	Low (device-dependent)	Low (sensitive to background and lighting)	Moderate (data-dependent)	High (trained on diverse datasets)
Long-term temporal awareness	No (single-event detection)	No (frame-based analysis)	Moderate (limited by vanishing gradient)	High (self-attention enables long-range motion learning)
Computational efficiency	High (low processing overhead)	High (simple feature extraction)	Low (sequential processing)	High (parallelized transformer architecture)
False positive rate	High (misclassification of normal activities)	High (threshold-based limitations)	Moderate (motion-based misclassification)	Low (adaptive deep learning model)
Suitability for edge deployment	Yes (lightweight devices)	Yes (low computational requirements)	No (requires extensive GPU resources)	Yes (optimized for real-time applications)

safety. By leveraging the hybrid deep learning model, falls can be detected accurately, even in cases where lighting conditions and occlusions pose challenges.

2) HEALTHCARE FACILITIES AND HOSPITALS

In hospital settings, fall incidents among patients, especially those recovering from surgery or mobility impairments, pose significant risks. The proposed system can be integrated into hospital surveillance infrastructure to continuously monitor patients without requiring them to wear additional devices. This non-intrusive approach ensures patient privacy while allowing medical staff to receive immediate alerts in case of an emergency, ultimately improving patient outcomes and reducing the burden on healthcare providers.

3) SMART HOMES AND INDEPENDENT LIVING

For elderly individuals who prefer to live independently, a home-based fall detection system is crucial for ensuring safety while maintaining autonomy. The proposed system can be deployed as part of smart home solutions, where it continuously analyzes video feeds from cameras installed in key locations such as bedrooms, bathrooms, and hallways. By integrating with IoT-based emergency response systems, caregivers or family members can be notified in real-time, allowing prompt intervention when needed.

4) WORKPLACE SAFETY AND INDUSTRIAL ENVIRONMENTS

Beyond healthcare applications, the system can also be utilized in workplace safety, particularly in industrial settings where employees risk falls due to hazardous conditions. For example, workers in construction sites, warehouses, and manufacturing plants often operate in environments where slips, trips, and falls are common. By implementing the proposed model in workplace monitoring systems, safety supervisors can proactively identify fall incidents and take necessary precautions to mitigate risks, ensuring compliance with occupational safety regulations.

5) PUBLIC SPACES AND SURVEILLANCE SYSTEMS

Another important application of the fall detection model is in public spaces, such as shopping malls, airports, and

transportation hubs. Large-scale surveillance systems can be enhanced with AI-powered fall detection to provide timely assistance to individuals who experience falls in high-footfall areas. This can be particularly beneficial in identifying medical emergencies, such as strokes or fainting incidents, enabling faster emergency response and potentially saving lives.

Discussing these real-world applications in depth highlights the proposed system's versatility and potential to contribute to various domains beyond healthcare. This expansion strengthens the research's motivation and demonstrates its broader societal impact.

This section has provided a detailed comparative analysis of fall detection techniques, highlighting their strengths and weaknesses. The proposed *hybrid YOLOv8 + TST model* overcomes the limitations of existing methods by combining high-precision object detection, long-range motion modelling, and computational efficiency. This ensures an optimal balance between real-time processing, accuracy, and generalizability, making it well-suited for healthcare and smart home applications.

VI. LIMITATIONS AND FUTURE WORKS

Despite the promising results achieved by our proposed fall detection system, certain limitations need to be acknowledged. These limitations primarily revolve around model robustness under diverse environmental conditions, privacy concerns, and the challenges posed by lighting variations. This section discusses these limitations in detail and outlines the future directions to address them.

A. MODEL ROBUSTNESS UNDER DIFFERENT ENVIRONMENTAL CONDITIONS

One key challenge in real-world fall detection systems is ensuring robustness under varied environmental conditions, including changes in lighting, occlusion, and variations in camera angles. While our current evaluation demonstrates high detection accuracy under controlled conditions, the model has not been extensively tested against these factors.

1) Lighting Variations:

- The performance of vision-based deep learning models can be significantly affected by lighting conditions. Low-light environments and sudden illumination changes may reduce feature visibility, leading to misclassifications or missed detections.
- Future work: To mitigate this, we plan to incorporate adaptive image enhancement techniques, such as histogram equalization and low-light image enhancement methods, to improve model performance in extreme lighting conditions. Training on an augmented dataset with synthetic lighting variations will also enhance the model's generalizability.

2) Occlusion Challenges:

- Partial occlusion, where furniture, walls, or other objects obstruct a person's body, poses another challenge. This can cause the model to fail to recognize the fall pattern correctly.
- Future work: We intend to explore multi-view camera setups or sensor fusion approaches integrating additional depth or infrared sensors to mitigate occlusion issues.

3) Camera Angles and Perspective Distortion:

- Variations in camera placement and viewing angles can affect the model's ability to detect falls consistently. A model trained on data captured from a single angle may struggle when deployed in environments where cameras are positioned differently.
- Future work: Future studies will include training on a diverse dataset with multiple camera perspectives to address this. The implementation of data augmentation techniques, such as geometric transformations and viewpoint adaptation, will also be explored.

Addressing these challenges will improve the robustness of the proposed fall detection model and ensure reliable operation across diverse environmental settings.

B. PRIVACY CONCERNS AND ETHICAL CONSIDERATIONS

Privacy remains a critical concern in vision-based fall detection systems, as they rely on continuous video monitoring of individuals in private environments. The deployment of AI-based surveillance technologies in healthcare and assisted living facilities raises ethical and regulatory challenges regarding data protection and user privacy.

1) Data Anonymization and Encryption Measures:

- To safeguard user privacy, all collected video data undergoes anonymization techniques, including face blurring and silhouette extraction, ensuring that personally identifiable information is not exposed.
- Additionally, all transmitted and stored data is encrypted using AES-256 encryption to prevent unauthorized access.

2) Edge Computing for Privacy-Preserving Fall Detection:

- Instead of transmitting video feeds to a cloud server for processing, on-device inference using edge computing is implemented. This ensures that raw video data remains local, reducing privacy risks associated with cloud-based surveillance.
- Future work: Future efforts will further optimize lightweight deep learning models to enable real-time processing directly on low-power edge devices, such as NVIDIA Jetson Nano or Raspberry Pi, enhancing privacy without compromising performance.

3) User Consent and Compliance with Ethical Guidelines:

- The system deployment strictly follows ethical guidelines, ensuring users provide explicit consent before collecting video data.
- Future work: Future work will explore privacy-preserving AI techniques, such as federated learning, where model training is performed locally on user devices without centralizing sensitive data.

By implementing these privacy-preserving measures, the proposed system ensures compliance with ethical standards while maintaining effectiveness in fall detection.

C. CHALLENGES POSED BY LIGHTING VARIATIONS AND MITIGATION STRATEGIES

Lighting variations are among the most significant environmental factors affecting vision-based AI models. Low-light conditions, shadows, glare, and sudden illumination shifts can lead to misclassifications or performance degradation. Addressing this challenge requires combining preprocessing techniques, robust model training, and sensor fusion approaches.

1) Preprocessing Techniques for Low-Light Adaptation:

- Traditional image enhancement techniques, such as histogram equalization and gamma correction, can improve contrast in low-light images, making fall-related features more distinguishable.
- Future work: We plan to integrate deep learning-based low-light enhancement models, such as Zero-DCE (Zero-Reference Deep Curve Estimation), to enhance images captured in extreme lighting conditions adaptively.

2) Augmenting Training Data for Lighting Robustness:

- A key limitation of current datasets is their lack of diversity in lighting conditions. Training on uniformly illuminated datasets may cause models to perform poorly in dimly lit or overexposed environments.
- Future work: We will expand our dataset to include images captured under variable lighting conditions, incorporating synthetically generated low-light and high-exposure variations using GAN-based data augmentation.

3) Multi-Sensor Fusion for Improved Reliability:

- RGB cameras alone may not be sufficient for robust fall detection in all environments. Complementary sensor modalities, such as infrared cameras, LiDAR, or depth sensors, can enhance model reliability.
- Future work: We aim to integrate thermal or depth-based sensors to supplement RGB-based vision models, ensuring stable performance across lighting conditions.

Implementing these strategies will make the proposed fall detection system more resilient to real-world challenges, improving its practicality in uncontrolled environments.

D. FUTURE RESEARCH DIRECTIONS

While this paper focuses on deep learning-based vision models, future research will explore the following directions:

- Multimodal approaches: Integrating vision-based models with wearable sensors (accelerometers, gyroscopes) for improved fall detection accuracy.
- Lightweight AI models for edge computing: Optimizing neural networks using pruning and quantization to enable real-time inference on resource-limited devices.
- Personalized fall detection models: Developing adaptive AI models that fine-tune detection strategies based on an individual's mobility patterns.
- Explainable AI for fall detection: Enhancing model interpretability by incorporating attention mechanisms highlighting the key areas contributing to fall classifications.
- Evaluating the model on additional real-world video recordings to further validate its robustness.

This section outlines the proposed fall detection system's primary limitations, focusing on model robustness, privacy concerns, and environmental challenges such as lighting variations. While our current approach demonstrates high detection accuracy, real-world deployments necessitate additional refinements to improve adaptability. The future research directions discussed here will ensure the system evolves into a more resilient, privacy-preserving, and computationally efficient fall detection framework suitable for diverse real-world applications.

VII. CONCLUSION

This paper introduces an innovative approach to fall detection by combining advanced object detection and temporal analysis techniques, specifically YOLOv8 and Time-Space Transformers. The primary motivation behind this study was to overcome the challenges associated with traditional fall detection systems, such as reliance on wearable sensors, which often result in discomfort, maintenance issues, and potential inaccuracies. In contrast, our image-based, non-intrusive solution enhances user comfort. It provides a scalable method for monitoring vulnerable individuals in real-time environments, such as homes, healthcare facilities,

and elderly care centres. The YOLOv8 model, known for its rapid and accurate object detection capabilities, was effectively leveraged to detect human figures in video sequences. By applying YOLOv8, the system can recognise humans in various positions, providing the spatial component of the fall detection system. However, fall detection requires more than object recognition; it also demands an understanding of temporal motion to distinguish between normal activities and falls. To achieve this, we incorporated Time-Space Transformers, which capture spatial features and temporal dependencies over time. This hybrid approach ensures that the system can accurately track the movement of individuals and identify sudden, abnormal changes in posture indicative of a fall. The evaluation of the proposed system on the CUCAFall and DiverseFALL10500 datasets demonstrated its superiority over existing state-of-the-art solutions. The system achieved higher accuracy with mAP scores of 0.9955 and 0.950, respectively, significantly reducing false positives and negatives, thus addressing a common issue in fall detection technologies. By integrating cutting-edge deep learning techniques, the proposed model improves detection accuracy and ensures real-time performance, which is critical for fall detection systems in practical applications. Moreover, this research highlights the potential of image-based fall detection systems to outperform wearable sensor-based approaches, particularly regarding user convenience and deployment scalability. The system's non-intrusive nature makes it adaptable to various environments without requiring direct user interaction or maintenance, enhancing its applicability in real-world scenarios. The ability to provide immediate alerts upon fall detection ensures timely interventions, which is crucial in preventing severe injuries, especially among elderly individuals.

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